ELSEVIER

Tectonophysics 299 (1998) 111–142

TECTONOPHYSICS

Cenozoic block rotation according to a conjugate shear system in central Europe — indications from palaeomagnetic measurements

U. Schreiber [a,*], S. Rotsch [b]

[a] *Department of Geology, University of Essen, Universitätsstrasse, 45117 Essen, Germany*
[b] *Institut of Geophysics, University of Münster, Corrensstrasse 24, 48149 Münster, Germany*

Received 1 October 1997; accepted 11 March 1998

Abstract

Results of palaeomagnetic measurements in the middle Tertiary volcanic provinces of Siebengebirge and Westerwald (northeastern Rhenish Massif, Germany) indicate that block rotation of the northeastern Rhenish Massif of the order of 10° to 16° has occurred since the Late Oligocene. Contemporaneous subsidence in the Upper Rhine graben, Neuwied basin, and Lower Rhine basin and intensive faulting in the Hessian depression document a new period of intraplate tectonism in central Europe. In the Tertiary, presumably from the Eocene to the Miocene, the axis of maximum horizontal stress rotated from the NNE–SSW to the NNW–SSE and the NW–SE direction, respectively. The new stress conditions may be responsible for the development of a conjugate shear system in central Europe north of the Alps. A model of a conjugate shear system is presented: a sinistral divergent strike-slip fault extends SSW–NNE from the Upper Rhine graben to the north. The fault splits up into two branches in the area of the Hessian depression. A conjugate dextral wrench fault system develops from the Bavarian Pfahl zone (southeastern Germany) to the Rhenish Massif. About 200 km to the northeast, a parallel dextral strike-slip fault is known from the southern Lower Saxony basin to the north of the Osning zone and Harz mountains. The northeastern block of the Rhenish Massif was created between dextral strike-slip faults. The block began clockwise rotation presumably in the Late Oligocene due to small changes in direction of the maximum horizontal compressive stress from NNW to NW and contemporaneous extension of the western European crust to the southwest. It is assumed that the block rotation is responsible for the V-shaped opening of the Lower Rhenish basin, and anticlockwise micro-block rotation within the Hessian depression. Opening between the rotating and nonrotating crust along the recent Rhine river in the central part of the Rhenish Massif formed triangular transrotational basins (basin of Neuwied) due to movements of crustal blocks in the western Rhenish Massif bounded by WNW–ESE dextral faults. Fault-plane solutions from recent earthquakes in the Rhine system support the model of block rotation and related crustal movements. Due to rotation of the northeastern Rhenish Massif the continuation of the Upper Rhine graben development to the north is prevented. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: palaeomagnetism; intraplate block rotation; conjugate shear system; Tertiary tectonics; central Europe

* Corresponding author. Tel.: +49 (201) 183-3100; Fax: +49 (201) 183-3101; E-mail: ulrich.schreiber@uni-essen.de

0040-1951/98/$ – see front matter © 1998 Elsevier Science B.V. All rights reserved.
PII: S0040-1951(98)00201-7

1. Introduction

The Cenozoic tectonism of central Europe is characterized by the development of large basins and graben systems at the same time as the uplift of crustal blocks. The Rhenish Massif with the surrounding graben systems, Upper Rhine graben, Lower Rhine basin and Hessian depression, is a key area for the reconstruction of Cenozoic tectonics in central Europe. In this area, several Tertiary and Quaternary volcanic fields have preserved the directions of the palaeomagnetic field (Nairn, 1962; Sherwood, 1990; Rotsch, 1997), which helps to identify rotation movements of crustal blocks. The variation in the intensity of tectonic activity is documented by the extent of sedimentary inputs in the grabens and basins. It is possible to determine the age of faults in the Cenozoic basins almost precisely whereas faults of the same tectonic event in the uplifted and deeply eroded areas are mostly not identifiable. As a result of the absence of stress indicators in unconsolidated sediments, palaeostress measurements in Tertiary clastics are not available. However, an analysis of brittle deformation in Mesozoic strata indicates the superposition of tectonic structures of different ages (Bergerat, 1987). Consequently, it is difficult or impossible to determine the age relations of deforming processes in the Mesozoic strata for the Cenozoic.

Although comprehensive data on regional geology, stratigraphy, tectonics and geophysics are available (for an introduction to the geology of the region, the reader is referred to Fuchs et al., 1983, Ziegler, 1992 and Meyer, 1994), the geodynamic process of the development of grabens and basins in central Europe in terms of the stress conditions in the crust is still poorly understood. The purpose of this paper is therefore to define the temporal and spatial relations between the basin and graben structures and the development of some volcanic fields on the basis of palaeomagnetic data.

2. Palaeomagnetic measurements

In the present study new palaeomagnetic results from Tertiary volcanic occurrences of the Siebengebirge and the Westerwald are presented. These volcanic fields are located at the northeastern part of the Rhenish Massif (Fig. 12). The new palaeomagnetic results are compared with (1) the apparent polar wander path (APWP) of Tertiary central Europe and — giving even more evidence — (2) with palaeomagnetic results of the neighbouring volcanic areas Hocheifel, Vogelsberg and Rhön, which were active at about the same time period. Older palaeomagnetic results from Siebengebirge and Westerwald derived from Nairn and Kürsten (1959) and Nairn (1962) did not seem to be very reliable because of missing detailed demagnetization experiments and corresponding analyses of remanent magnetization. Here we present the results of our new palaeomagnetic investigations of 36 sites of the Westerwald and 19 sites of the Siebengebirge Tertiary volcanic fields (Rotsch, 1997; Fig. 1, Table 1).

2.1. Sampling and laboratory techniques

In order to determine reliable palaeodirections from volcanic fields of the Siebengebirge and the Westerwald, only those sites were sampled for which geological setting, expansion of the site and rock magnetic properties seemed to promise successful palaeomagnetic investigations. Sites with rock material showing obvious outer signs of weathering have been rejected. Accordingly, samples were mainly taken from quarries with fresh rock material outcrops. To get an overview of the palaeodirections in both volcanic fields, it was tried to sample as many sites distributed all over the whole volcanic fields as possible. Furthermore, sampling was focused especially on isotopically dated sites. This enabled us to compare the recorded palaeodirections from the Siebengebirge and the Westerwald with results from surrounding volcanic regions and with the APWP compilations for Tertiary central Europe. Isotopic dating (whole rock dating and $^{40}Ar/^{39}Ar$ mineral dating) of 21 sites in the Siebengebirge and 19 sites in the Westerwald by Lippolt and Todt (1978) and Todt and Lippolt (1980) shows, that the main period of Tertiary volcanic activity for both volcanic fields can be dated back to the time span of 22–30 Ma. However, in some cases isotopic ages were found to be in contradiction to geological facts (e.g. Schreiber, 1996). Due to different reasons, unfortunately, not all of these dated sites met the conditions

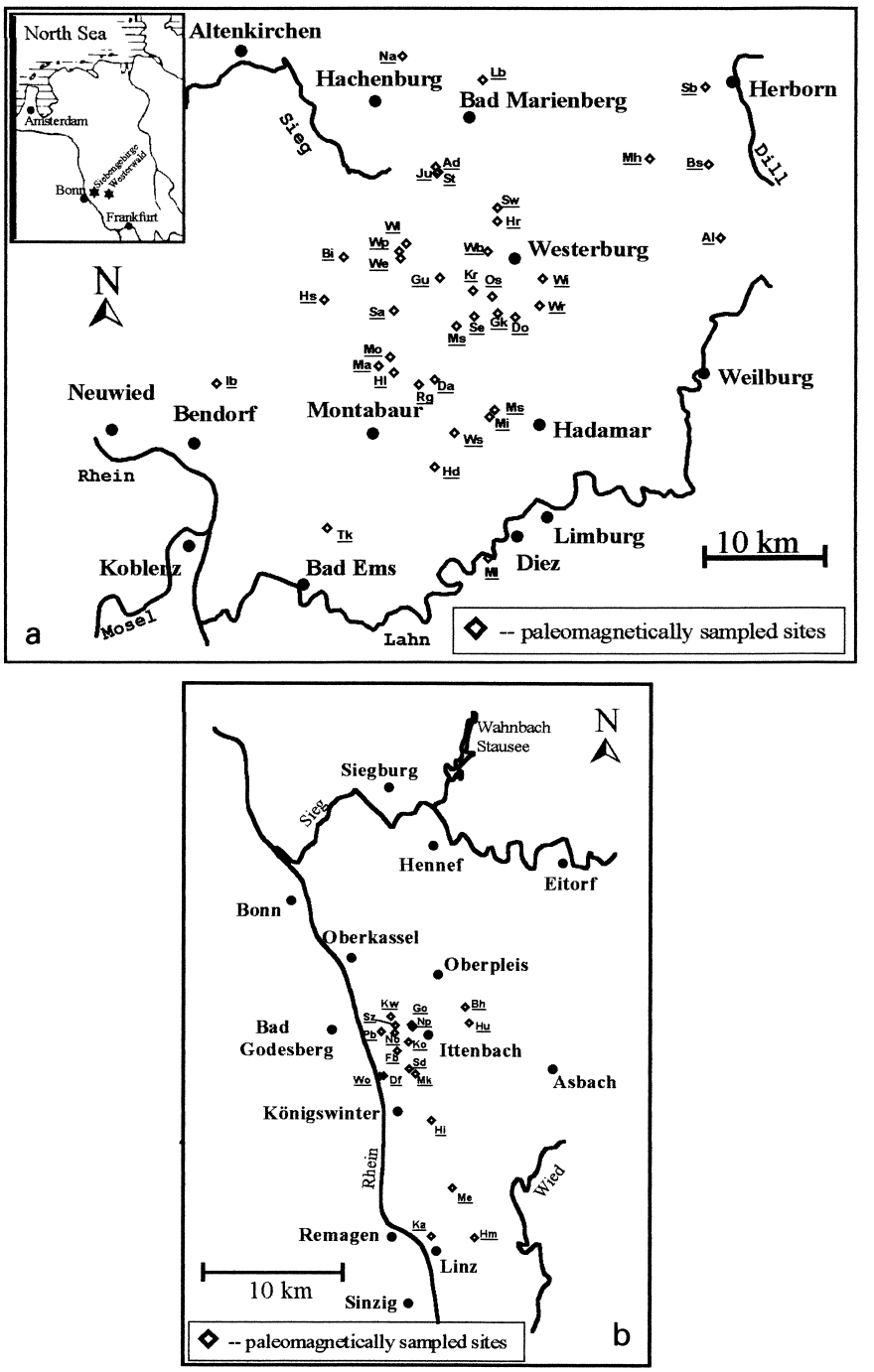


Fig. 1. (a) Palaeomagnetically investigated occurrences of the Tertiary Westerwald volcanic field. (b) Palaeomagnetically investigated occurrences of the Tertiary Siebengebirge volcanic field. For definition of site name abbreviations see Table 1; also see Fig. 12.

Table 1
Palaeomagnetic sites from Tertiary volcanic occurrences of the Westerwald and the Siebengebirge

Abbr.	Site	r	h	Map	Pc
Westerwald					
Ad	Stöffel-Adrian	3421725	5608750	5313	Bs
Ju	Stöffel Urmacher	3421600	5609200	5313	Bs
St	Stöffel-Stockum	3421800	5608850	5313	Bs
Al	Allendorf	3448125	5604000	5415	Bs
Bi	Bittersberg	3413000	5602575	5412	Bs
Bs	Beilstein	3447000	5609400	5315	Bs
Da	Dahlen	3421500	5593475	5513	Tr
Do	Dorndorf	3429000	5598100	5413/14	Bs
Gk	Girkenroth	3427425	5598400	5413	Bs
Gu	Guckheim	3425075	5600075	5413	Bs
Hd	Heilberscheid	3421525	5587000	5513	Bn
Hl	Hülsberg	3416300	5594500	5512	At
Hr	Hergenroth	3427350	5605225	5413	Bs
Hs	Hahn (bei Selters)	3411200	5599425	5412	Lt
Ib	Isenburg	3401200	5593200	5511	Tr
Kr	Kranstein	3426900	5599675	5413	Bs
Lb	Lautzenbrücken	3426000	5615650	5313	Bs
Ma	Mahlberg	3417325	5595175	5512/13	Ba
Mb	Ölberg (WW)	3427125	5591275	5513	Bs
Mh	Münchhausen	3441525	5609825	5315	Bs
Mi	Miltensberg	3426675	5590750	5513	Bs
Mo	Moschheim	3417725	5594000	5513	Ta
Ms	Markstein	3423550	5597475	5413	Bs
Na	Nauberg	3418475	5617400	5313	Ba
Os	Obersain	3422000	5601025	5413	Ha
Rg	Ruppach-Goldhausen	3420050	5593087	5513	La
Sa	Sainerholz	3417700	5598575	5413	Ba
Sb	Steinringsberg	3446725	5615125	5315	Bs
Se	Sengelberg	3425225	5598150	5413	Hba
Sw	Stahlhofen/Wiesensee	3427350	5606225	5413	Bs
Tk	Teufelsköpfe	3411500	5582500	5612	Bs
Wb	Westerburg	3426450	5603000	5413	Bs
We	Weidenhahn	3418275	5602475	5413	Qt
Wi	Wickertsberg	3431600	5601000	5414	Hob
Wl	Wölferlingen	3418825	5603575	5413	Lt
Wp	Wölfers Berg	3418175	5603000	5413	Hb
Wr	Wilsenroth	3431300	5599000	5414	Ba
Ws	Wellerstein	3423350	5589550	5513	Bs
Siebengebirge					
Bh	Berghausen	2592925	5618875	5309	Bs
DF	Drachenfels	2585475	5615075	5309	Qt
Fb	Froschberg	2586950	5616450	5309	Tr
Go1	Großer Ölberg 1	2588200	5617900	5309	Ab
Go2	Großer Ölberg 2	2588300	5617875	5309	Tr
Hi	Himmerich	2590000	5612675	5309	Bs
Hm	Hummelsberg	2593725	5606300	5409	Bs
Hu	Hühnerberg	2593250	5618000	5309	Bs
Ka	Kasbach-Ohlenberg	2590000	5606350	5409	Bs
Ko	Kleiner Ölberg	2588275	5617750	5309	Ab
Kw	Kleiner Weilberg	2586400	5618350	5309	Ab
Me	Meerberg	2591825	5609000	5309	Bs

Table 1 (continued)

Abbr.	Site	r	h	Map	Pc
Mk	Merkenshöhe	2588575	5615230	5309	Tr
No	Nonnenstromberg	2586725	5617425	5309	Ab
Np	Nasseplatz	2587975	5616950	5309	Tr
Pb	Petersberg	2585550	5617500	5309	Ab
Sd	Schulz-Denkmal	2588000	5615500	5309	Tr
Sz	Stenzelberg	2586825	5617875	5309	Ql
Wo	Wolkenburg	2585800	5615100	5309	Ql

Columns: abbreviation of site; name of site; coordinates; number of topographic map; Pc, petrographic composition. Bs = basalt; Ba = basanite; Ab = alkali basalt; Ha = hawaiite; Hob = basalt with amphibole and olivine phenocrysts; Hb = amphibole-bearing basalt; Tr = trachyte; Bn = benmoreite; At = alkali-rich trachyte; Lt = silica-rich trachyte; Ta = trachyandesite; La = latite; Hba = amphibole-bearing andesite; Qt = quartz-rich trachyte; Ql = quartz-rich latite.

for successful palaeomagnetic investigations. Along each site at least ten locally well distributed cores were drilled using an electrically powered drill with a drillholder and a diamond bit. The cores were oriented by magnetic compass and, if possible, with a sun compass and cut into several standard specimens of 1 inch. The slices in between were used for rock magnetic investigations. Alternating field (AF) demagnetization and thermal demagnetization were performed with rather the same equipment described by Schnepp (1994) (Digico and Molspin spinner magnetometer, self-built demagnetization coil and oven, horizontal magnetic balance for thermomagnetic measurements and self-built impulse magnetizer for isothermal remanent acquisition (IRM) in maximum fields up to 1.4 T and Molspin vibrating sample magnetometer for apparent hysteresis).

2.2. Rock magnetic properties

To identify the remanence carrying mineral fraction, thermomagnetic measurements of several samples of each site were performed and the Curie temperatures were determined. Results for the 55 sites may be summarized as follows. The material can roughly be divided into two groups: LT1-3 (more than 60% of the sites) with low Curie temperatures below 300ºC, and HT1-5 with Curie temperatures between 300ºC and 650ºC (Fig. 2). Samples of the first group are supposed to contain unexsolved titanium-rich titanomagnetite (TM40–TM75). Samples of the second group indicate high-temperature exsolved titanomagnetite (down to TM10), for which a titanium-poor titanomagnetite remains as remanence carrying mineral fraction. Apart from three exceptions (Fig. 2, LT3) the thermomagnetic curves do not show any serious indication of low-temperature oxidation processes such as maghemitization. Rock magnetic parameters derived from hysteresis and remanent hysteresis loops are used to estimate the grain size of the remanence carrying mineral fraction. The diagram after Day et al. (1977) yields predominantly pseudo-multidomain and multidomain particles with diameters between 1 and 40 μm (Fig. 3). Furthermore, remanent hysteresis loops indicated that the remanent magnetization of all measured samples yielded more than 90% of the remanent saturation magnetization already at 300 mT (Fig. 4). So it can be concluded, that high-coercive minerals like hematite have no serious influence on the bulk rock magnetic behaviour of the Siebengebirge and Westerwald samples. Reflected light microscopy of polished sample surfaces also verified these results. Only a few samples showed very weak indications of high-temperature oxidation processes (exsolvation lamellas), which presumably took place during rock cooling. This might have caused chemical remanent magnetization components, so that for the resulting primary magnetization one has to expect primary chemical remanent magnetization components as well as primary thermoremanent components. Experiments concerning acquisition rates of viscous remanent magnetization (in constant magnetic field of 50 μT, 15 days acquisition time) showed similar results as those derived for ocean basalts (e.g. Lowrie and Kent, 1976, 1978), samples from the Steens mountains or the Massif Central (southern France, Prévot, 1981). VRMs are supposed

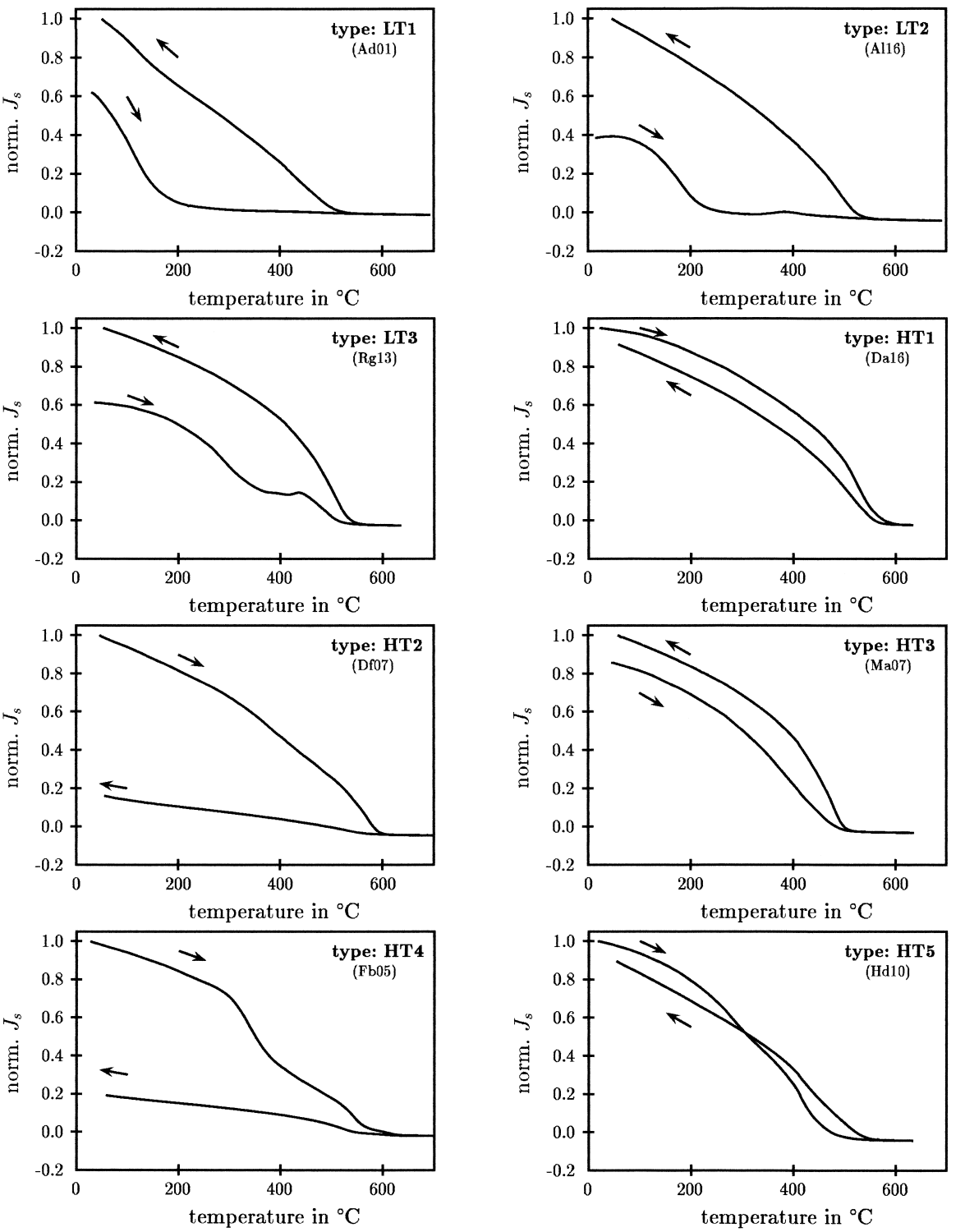


Fig. 2. Overview of types of thermomagnetic curves observed in Tertiary volcanic rocks from Siebengebirge and Westerwald. *LT* = low-temperature type; *HT* = high-temperature types. In brackets: name of sample.

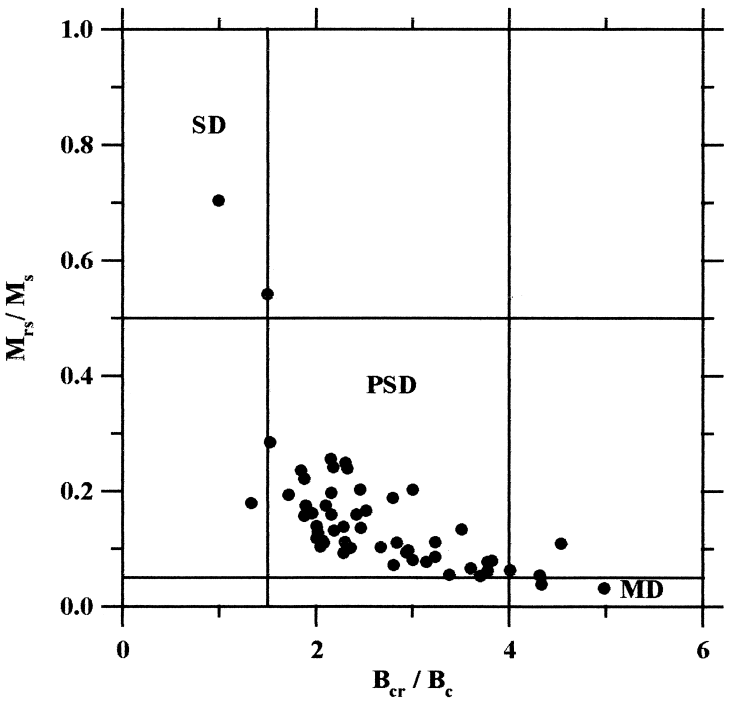


Fig. 3. Diagram after Day et al. (1977) for rock samples from Siebengebirge and Westerwald. Predominantly pseudo-multidomain and multidomain particles with diameters between 1 and 40 μm are observed.

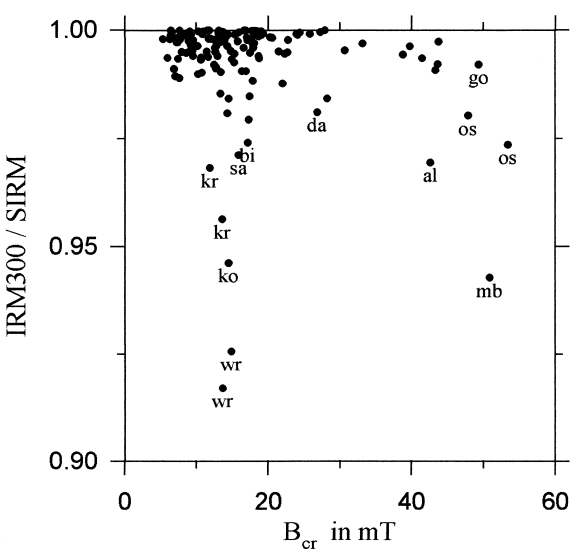


Fig. 4. Saturation behaviour of Siebengebirge and Westerwald rock samples: the relation between isothermal remanent magnetization at 300 mT (IRM300) and remanent saturation magnetization (SIRM) as a function of remanent coercive force B_{cr}.

to be the main reason for secondary overprints of the primary remanence components.

2.3. NRM and demagnetization experiments

The direction of the natural remanent magnetization (NRM) typically shows considerable dispersion within most of the sites. This might be related to secondary remanence overprints mainly caused by viscous remanence acquisition, which seemed to be predominantly unsystematic. Demagnetization experiments were performed with at least ten samples per site to eliminate secondary remanent magnetization components. Because of magnetomineralogical instability during heating steps at temperatures of about 200°C and above, thermal demagnetization experiments failed in most cases. Fortunately, the AF demagnetization method was successful for samples from all sites. In a maximum AF peak field of 260 kA/m, all samples could be demagnetized to less than 10% of their original remanence. This confirms that mineral fractions with high-coercive

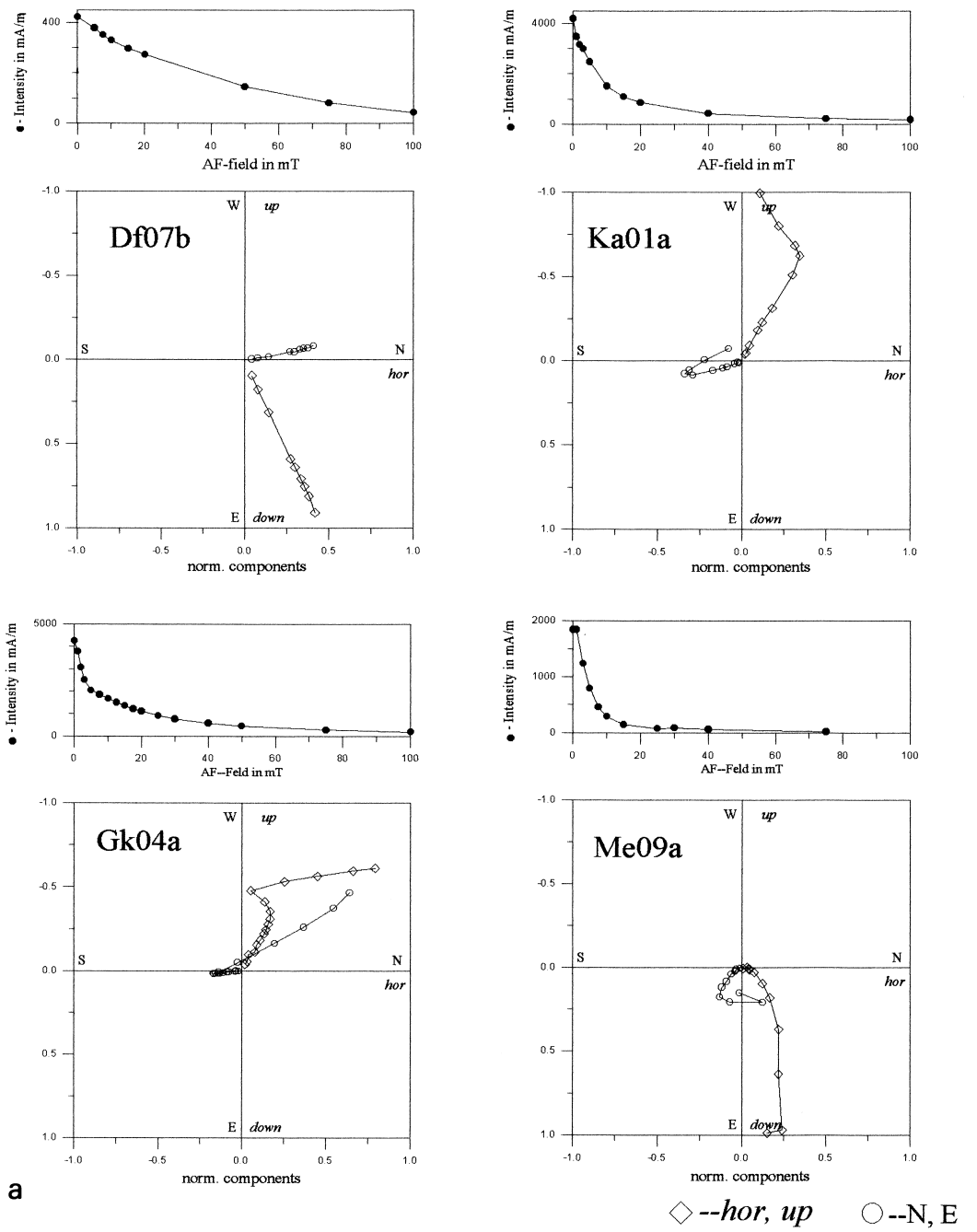


Fig. 5. (a) Some typical examples for AF demagnetizations. For each sample the top diagram shows the course of remanent magnetization intensity as a function of applied AF field, below the Zijderveld diagram is given.

forces like hematite do not seriously affect the samples from Siebengebirge and Westerwald. Fig. 5 gives some representative examples of Zijderveld diagrams from AF demagnetization experiments. Sample Df07b (Fig. 5a) represents the scarce group (about 10%) of one-component systems and is on

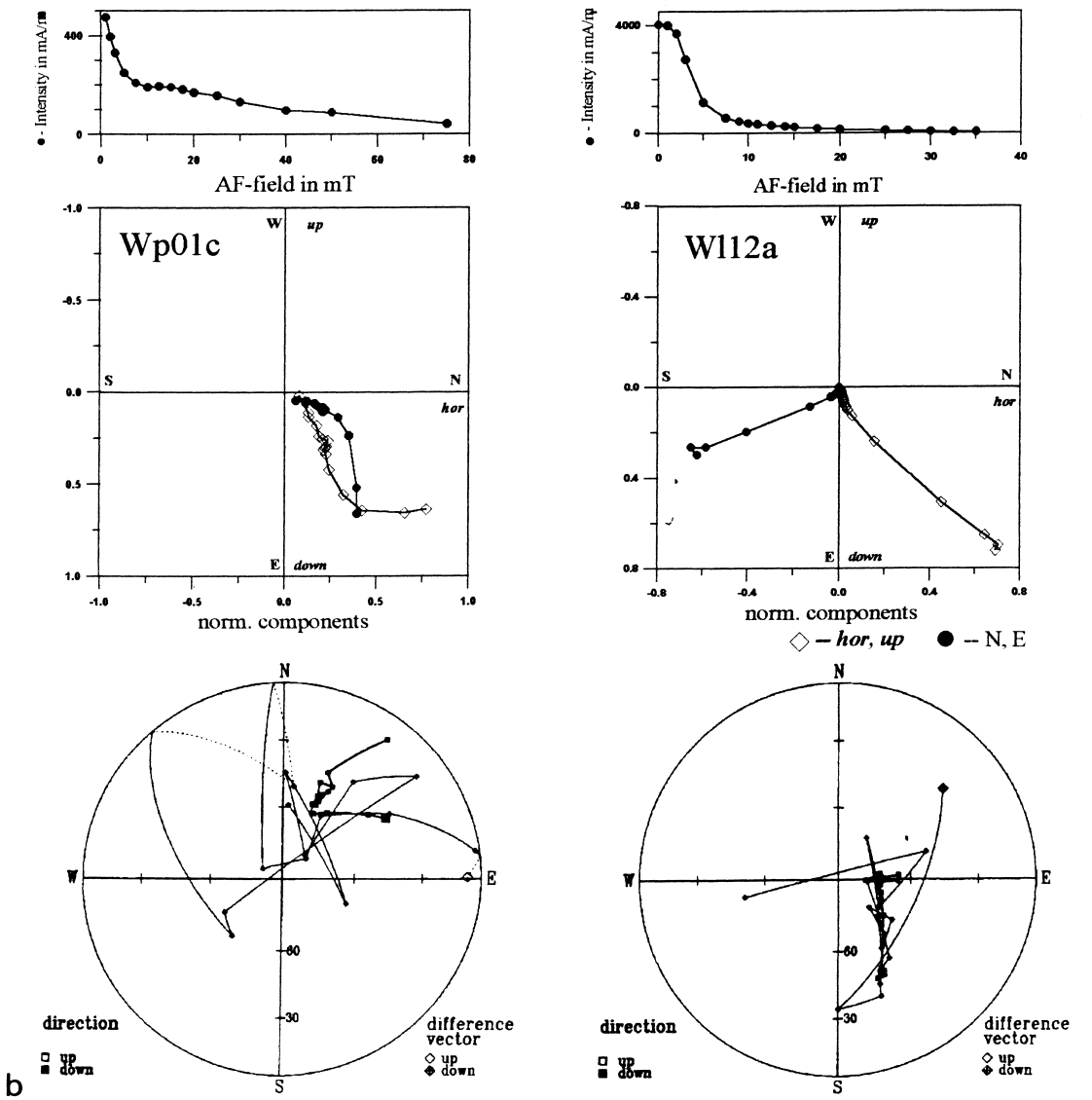


Fig. 5 (continued). (b) Two further examples for AF demagnetizations. Presentation same as (a). In addition, the equal-area projection diagrams at the bottom show the course of remanence direction vectors and difference vectors.

the whole free of any secondary magnetic overprints. Typically, the remanent magnetization of the Siebengebirge and Westerwald samples consists of multi-component systems. Coercivity spectra for the individual components were found to be distinctively different in about 40% of the samples (e.g. sample Ka01a, Fig. 5a) but in about 50% of all cases they showed more or less overlap with each other (e.g. samples Gk04a, Me09a, Fig. 5a). Zijderveld diagrams and differential-vector analyses were suitable to determine the characteristic remanent magnetization (ChRM) directions for samples Df07b or Ka01a. In cases like sample Gk04a or Me09a it became more and more difficult to get reliable results, and for Wp01c or Wl12a (Fig. 5b) it was even impossible.

Table 2
Palaeomagnetic results from Westerwald and Siebengebirge

Site	ChRM					VGP			
	Method	*N*	Decl.	Incl.	α_{95}	Plat.	Plong.	dp	dm
Westerwald									
Ad	see St	–	–	–	–	–	–	–	–
Ju	see St	–	–	–	–	–	–	–	–
St	(+ Ad + Ju) p	13	269.7	−32.4	8.0	−13.7	266.7	5.1	9.1
Al	lp	8	357.7	54.5	10.1	74.4	195.3	10.1	14.3
Bi*	(l)	13	155.9	−58.5	8.6	−69.6	73.3	9.5	12.8
Bs*	(l)	5	22.9	52.8	6.8	65.9	135.4	6.5	9.5
Da	p	7	255.5	−72.3	8.3	−47.4	238.2	13.1	14.8
Do	lp	8	191.7	−42.3	5.0	−62.4	344.5	3.8	6.2
Gk	l	8	185.4	−65.8	3.7	−85.7	310.9	4.9	6.0
Gu	lp	7	228.0	−62.0	6.9	−57.0	271.7	8.3	10.7
Hd	p	7	10.4	51.2	14.6	69.9	161.4	13.4	19.8
Hl	p	6	30.2	38.7	4.8	52.9	137.1	3.4	5.7
Hr	lp	8	202.8	−59.8	7.1	−71.3	301.3	8.1	10.7
Hs	l	7	61.8	34.6	6.7	32.4	107.1	4.4	7.7
Ib	lp	8	26.1	56.7	7.8	67.2	122.9	8.2	11.3
Kr	l	8	18.2	73.0	4.3	76.8	53.6	6.9	7.7
Lb	p	7	204.6	−54.6	8.9	−66.4	309.7	8.9	12.6
Ma*	(p)	3	–	–	–	–	–	–	–
Mb	lp	9	208.6	−72.2	7.7	−72.0	244.5	12.1	13.7
Mh*	(l)	2	–	–	–	–	–	–	–
Mi	p	6	3.8	72.1	5.4	82.9	25.0	8.4	9.5
Mo	lp	6	195.8	−74.6	8.7	−72.6	240.0	14.4	15.8
Ms	l	6	202.4	−50.3	6.7	−64.4	318.8	6.0	9.0
Na	l	9	198.5	−49.0	4.2	−65.1	327.0	3.6	5.5
Os	l	10	207.4	−74.7	4.8	−71.5	231.8	8.0	8.8
Rg	p	8	287.9	78.9	5.7	52.1	333.4	10.2	10.7
Sa*	(l)	1	–	–	–	–	–	–	–
Sb	lp	6	151.8	−71.8	8.9	−66.5	138.8	13.8	15.7
Se	l	8	206.5	−64.3	4.0	−72.0	281.6	5.1	6.4
Sw*	(l)	3	–	–	–	–	–	–	–
Tk*	(l)	5	345.0	71.3	6.1	79.5	315,1	9.3	10.7
Wb	p	7	174.7	−29.2	4.8	−54.8	16.8	2.9	5.3
We	p	7	5.3	40.3	4.5	62.1	177.4	3.3	5.4
Wi	p	6	1.6	71.2	5.1	84.7	17.8	7.7	8.8
Wl	lp	6	19.9	54.1	4.1	68.5	138.0	4.0	5.7
Wp	lp	6	348.2	59.0	7.2	76.4	229.9	7.9	10.7
Wr	l	9	252.1	−73.4	2.2	−49.7	236.9	3.6	4.0
Ws	l	7	62.3	62.5	3.1	48.3	82.0	3.8	4.9
Siebengebirge									
Bh	lp	8	233.2	−44.0	7.4	−42.7	288.5	5.8	9.2
Df	l	8	344.6	65.7	2.3	79.6	268.3	3.0	3.7
Fb	l	9	2.9	62.3	3.0	82.6	170.6	3.6	4.7
Go1	l	7	216.8	−57.7	1.9	−61.3	288.7	2.1	2.8
Go2	l	10	197.7	−59.9	2.9	−74.2	309.1	3.3	4.4
Hi	lp	10	8.2	40.5	7.4	61.8	171.2	5.4	8.9
Hm	l	8	175.9	−56.4	7.4	−76.0	21.1	7.7	10.7
Hu	p	8	248.9	−53.6	7.4	−39.3	273.1	7.2	10.4
Ka	l	10	160.4	−62.8	2.3	−75.4	79.3	2.9	3.6
Ko	l	6	20.0	67.5	5.0	77.3	91.0	7.0	8.3
Kw*	(l)	4	209.7	−59.8	7.3	−67.2	291.6	8.4	11.1

Table 2 (continued)

Site	ChRM					VGP			
	Method	*N*	Decl.	Incl.	α_{95}	Plat.	Plong.	dp	dm
Me	l	6	140.0	19.5	3.9	−20.1	49.7	2.1	4.0
Mk	l	6	350.3	64.6	3.5	82.3	248.1	4.5	5.7
No	l	10	9.9	59.9	5.1	77.9	148.6	5.8	7.7
Np	p	7	46.3	−10.4	3.3	21.4	136.6	1.7	3.3
Pb	p	6	338.3	−75.3	4.5	−24.2	198.1	7.5	8.2
Sd	p	6	18.5	50.8	7.4	66.4	144.6	6.8	10.0
Sz	l	10	29.8	65.4	3.9	70.4	94.9	5.1	6.2
Wo	l	11	4.4	67.7	3.0	87.2	96.9	4.2	5.1

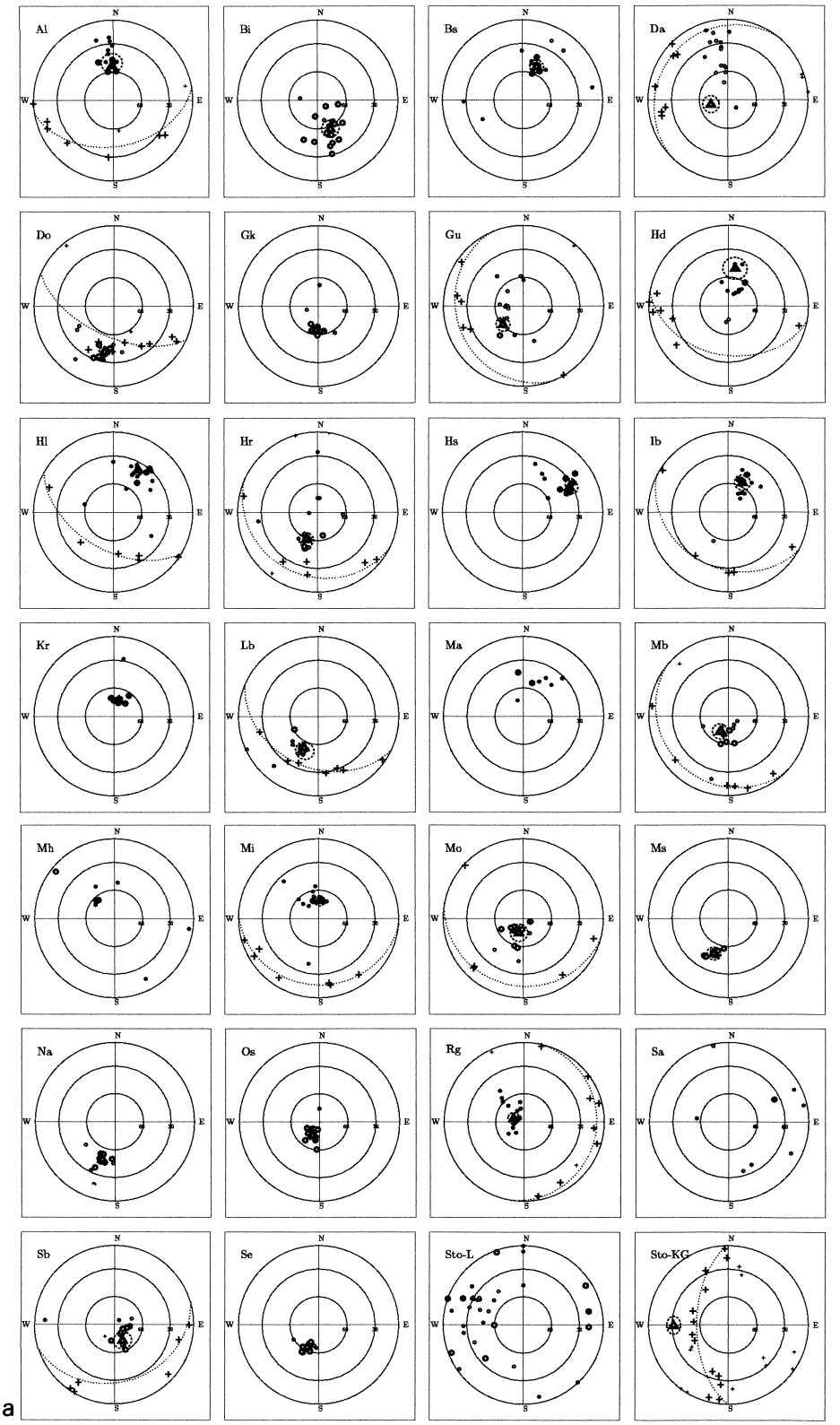
Columns: site, method to determine ChRM directions (l = linear fits using PCA; p = planar fits with PCA; lp = combination of l and p; see Section 2.4); *N*, number of rock samples for which ChRM directions were determined successfully; Decl., Incl., declination and inclination of mean ChRMdirection; α_{95}, precession parameter and error cone of the Fisher statistics; Plat., Plong., latitude and longitude of the VGP with dp, dm, half axis of error ellipse. Asterisk at site abbr., no reliable ChRM direction could be determined (tried method in brackets). These directions were eliminated form further calculations.

2.4. Determination of ChRM

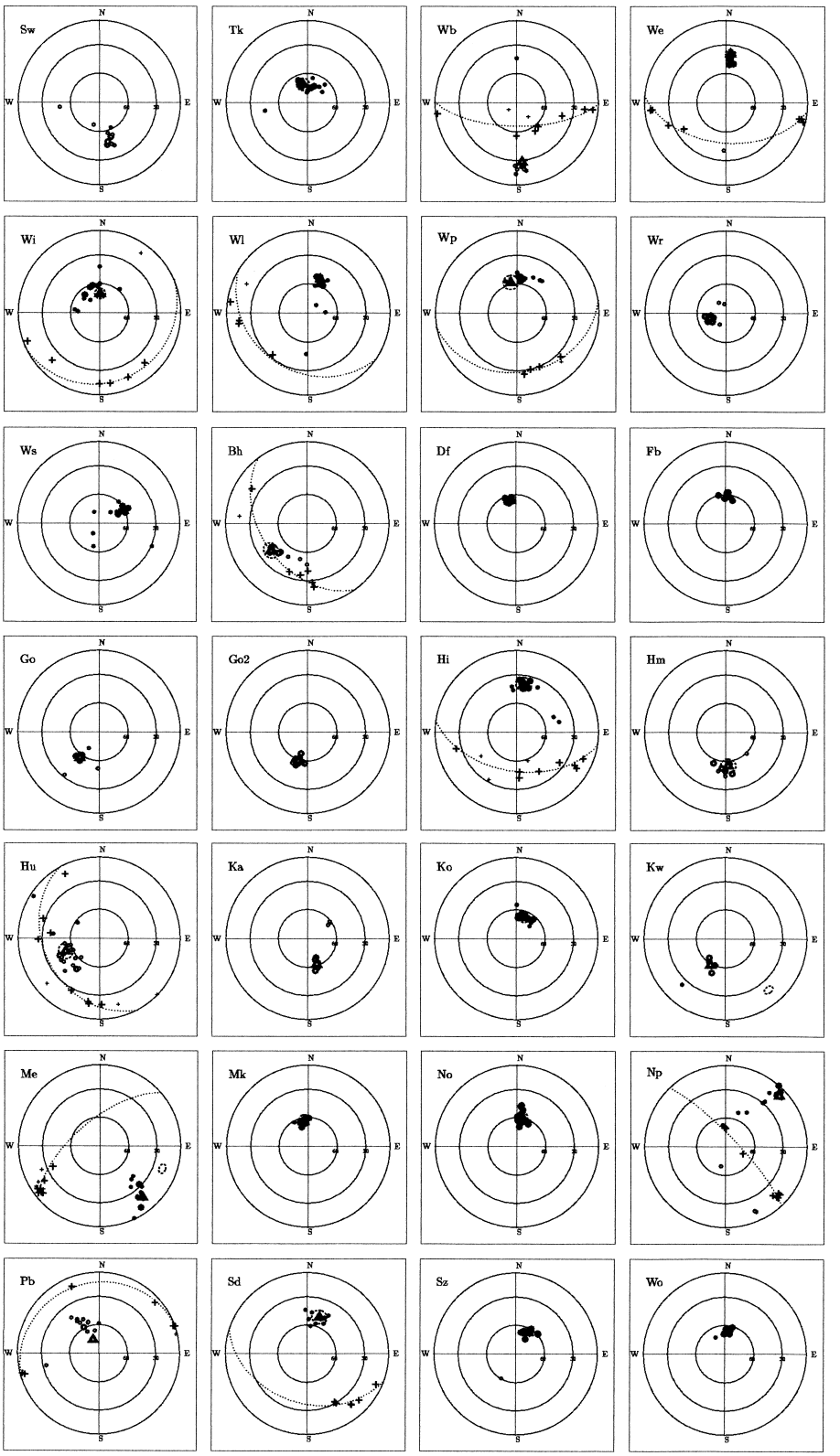
As demonstrated in the preceding section, in this study conventional analyses of Zijderveld diagrams and differential-vector analyses are only suitable for the demagnetization of one-component or simple multi-component remanent magnetization systems. Otherwise, these methods do not lead to well defined reliable directions of the ChRM because of overlapping coercivity spectra. For this reason a principal component analysis (PCA) based on the algorithm of Kirschvink (1980) was applied to analyse the demagnetization paths of all 700 investigated rock samples. In a first step, a PCA was used to fit demagnetization paths with linear segments (LS). Following the algorithm of Kirschvink (1980), the maximum angular deviations (MAD) of the fitted LS were calculated. In addition, the projections in three orthogonal planes (e.g. the *xy*-, the *zx*- and the *zy*-plane) of the points belonging to each LS were fitted also by free lines. Between each of these free lines and the corresponding projection of the LS (determined with PCA) the angle α was calculated. Finally, the ChRM segment could be identified as the segment with the least maximum angle α between free lines and corresponding origin-passing projections of LS. To decide whether the direction given by the ChRM segment can be taken as the primary direction of the rock sample, two empirically established criteria have to be met: (1) the MAD and (2) also the maximum angle α should be below 3º (Rotsch, 1997). Unfortunately, about 25% of the samples show coercivity spectra with a rather strong overlap of the individual components, so that no well defined ChRM segments could be determined in using linear fits from PCA. In these cases demagnetization paths were fitted with planes from PCA. The plane, which contained the origin, was identified as the ChRM-direction-containing plane. Again, the MAD was calculated after Kirschvink (1980) and the fit was accepted, if the MAD was lower than 5º.

2.5. Calculation of site-mean directions

In order to calculate the site-mean directions, we distinguished three cases: (1) Sample-ChRM directions were successfully determined with linear fits of the sample demagnetization paths (using the PCA method) for at least six samples of the site. (2) Linear fits of sample demagnetization paths failed to determine a ChRM segment but for at least six samples ChRM-containing planes were found with PCA. (3) Demagnetization paths could be fitted with planes of only four or five samples and ChRM-containing planes were successfully determined. In addition, for further samples of the site ChRM directions were successfully determined in using linear fits from PCA. For the different cases, site-mean directions were determined as follows. In case (1) (20 sites), site-mean directions and confidence angles α_{95} were calculated with the Fisher statistics (Fisher, 1953). In the second case (15 sites), the mean ChRM direc-


Al
Bi
Bs
Da
Do
Gk
Gu
Hd
Hl
Hr
Hs
Ib
Kr
Lb
Ma
Mb
Mh
Mi
Mo
Ms
Na
Os
Rg
Sa
Sb
Se
Sto-L
Sto-KG
N
W
E
S

a


Sw
Tk
Wb
We
Wi
Wl
Wp
Wr
Ws
Bh
Df
Fb
Go
Go2
Hi
Hm
Hu
Ka
Ko
Kw
Me
Mk
No
Np
Pb
Sd
Sz
Wo
N
W
E
S

b

tions were calculated using the method of converging remagnetization circles (Halls, 1978). In the third case (12 sites), (a) the mean ChRM direction from linear fitted demagnetization paths was determined, and (b) the mean ChRM direction from planar fitted demagnetization paths was estimated using the method of converging remagnetization circles. Then, referring to the amount of samples, the site-mean direction was calculated as a weighting mean from (a) and (b). Only site means based on at least six rock samples (locally well distributed along the sampled site) were accepted, results yielding dispersion of more than $\alpha_{95} = 15^{o}$ (or $k < 80$) in Fisher statistics were rejected. Unfortunately, for eight sites no reliable site-mean directions could be determined that fulfilled the required conditions. For all site-mean directions virtual magnetic poles (VGP) were calculated. Table 2 and Fig. 6 give an overview of site-mean directions and ChRM-direction distributions of all sites.

2.6. The new Tertiary palaeopole of Siebengebirge and Westerwald

Altogether, ChRM directions and related VGPs were determined for 18 sites of the Siebengebirge and 29 sites of the Westerwald (Fig. 7). Palaeopoles for each volcanic area have been calculated. To exclude intermediate VGP, the cut-off method described by Vandamme (1994) was used. As a result of subsequently applied F-statistics, these palaeopoles of Siebengebirge and Westerwald indicated not to be significantly different from each other. Furthermore, isotopic dating indicates simultaneous volcanic activity in Siebengebirge and Westerwald (Fig. 8). Consequently, ChRM directions and related VGPs of both volcanic areas were put together and a common palaeopole was computed with the cut-off method of Vandamme (1994). Statistical parameters of this common VGP distribution were in good agreement with data given in literature. The angular standard deviation amounts to 21.6º (20º, McFadden et al., 1991), and the cut-off angle determined with the method of Vandamme (1994) was 43.8º (40º, Wilson et al., 1972; 45º, Watkins, 1973). The combined data set of Siebengebirge and Westerwald results in a palaeopole position of 119.2º longitude and 75.3º latitude. The related palaeodirection is of 19.2º declination and 60.8º inclination. The result of a reversal test was positive (Rotsch, 1997).

To estimate the time span the new palaeopole can be related to, one has to refer to the results of isotopic dating. Of the 47 sites, for which VGP could be determined, 16 sites have been isotopically dated by Lippolt and Todt (1978) and Todt and Lippolt (1980). As mentioned in Section 2.1, the precision of these ages is under discussion. Nevertheless, they are suitable to fix the time window of the whole eruption period of the Tertiary Siebengebirge and Westerwald to the time interval lasting from about 18 to 30 Ma (Fig. 8).

2.7. Discussion of the new Tertiary palaeopole

Fig. 9 gives an overview of the most important APWPs of Tertiary central Europe. While all compilations are in rather good agreement at about 50 Ma, especially large discrepancies are observed in the 0–40 Ma time period and no *standard path* can be defined at all. Three reasons are supposed to be most important for these discrepancies: (1) The fairly unsuitable database for this time span. (2) Different compilation methods which have especially strong influence on time spans with only a few data points. (3) Differing data bases are used, only Van der Voo (1990) and Schott et al. (1994) refer to the same data base. Taking also into account the 95% circles of confidence of the APWP points, only limited conclusions might be drawn, if the new Tertiary palaeopole of Siebengebirge and Westerwald is compared with the APWP compilations. Nevertheless, in all cases the position of

Fig. 6. (a, b) Distribution of ChRM directions from all sites. For definition of site name abbreviations see Table 1. *Al-Ws* = sites from the Westerwald; *Bh-Wo* = sites from the Siebengebirge. Solid symbols, normal directions; open symbols, inverse directions; small symbols, rejected sample-ChRMs; crosses, normal vectors to remagnetization planes; small crosses, rejected normal vectors to remagnetization planes; dotted curves, fitted great circles to normal vectors of remagnetization planes; triangle, site-mean ChRM with dashed α_{95} confidence circles.

the Siebenbirge/Westerwald palaeopole would be expected at a higher latitude and a more eastern longitude (Fig. 9). In order to explain this deviation, we compared the new palaeopole position with palaeomagnetic results of the neighbouring volcanic areas: (a) the older (24–46 Ma) Hocheifel volcanics (Weinreich and Bleil, 1984; Schnepp and Boyne, 1996); (b) the younger (15–18 Ma) Vogelsberg volcanic occurrences (Sherwood, 1990); and (c) the Rhön volcanics (Bock and Soffel, 1971), which were active in the same time period (19–22 Ma), but because of less reliable quality, these data have to be interpreted carefully (the data set obviously does not pass a reversal test). Palaeomagnetic results from other Tertiary volcanic fields of central Europe are not considered here, because they are of even lower quality (see e.g. the review given by Weinreich and Bleil, 1984). The most striking fact obvious in Fig. 10 is that palaeopoles and palaeodirections from Hocheifel, Vogelsberg and Rhön are in good agreement to each other and also fairly match the APWPs, but are significantly different from the results of Siebengebirge and Westerwald. As can be seen in Fig. 10, only the inclination of the palaeodirection of Siebengebirge and Westerwald is observed in the expected range. In our opinion the remarkable deviation of the palaeodeclination of Siebengebirge and Westerwald is a serious indication for a tectonic rotation. Interpreting our data from this point of view leads to the supposition that the northeastern part of the Rhenish Massif, including the Siebengebirge and Westerwald volcanic fields, or the Siebengebirge and Westerwald as discrete blocks might have undergone a block rotation. The amount of clockwise rotation from palaeomagnetic measurements is estimated at 16º (Fig. 10).

3. Other indications of crustal rotation

Apart from palaeomagnetic results, the following arguments indicate a Tertiary rotational history for the northeastern Rhenish Massif.

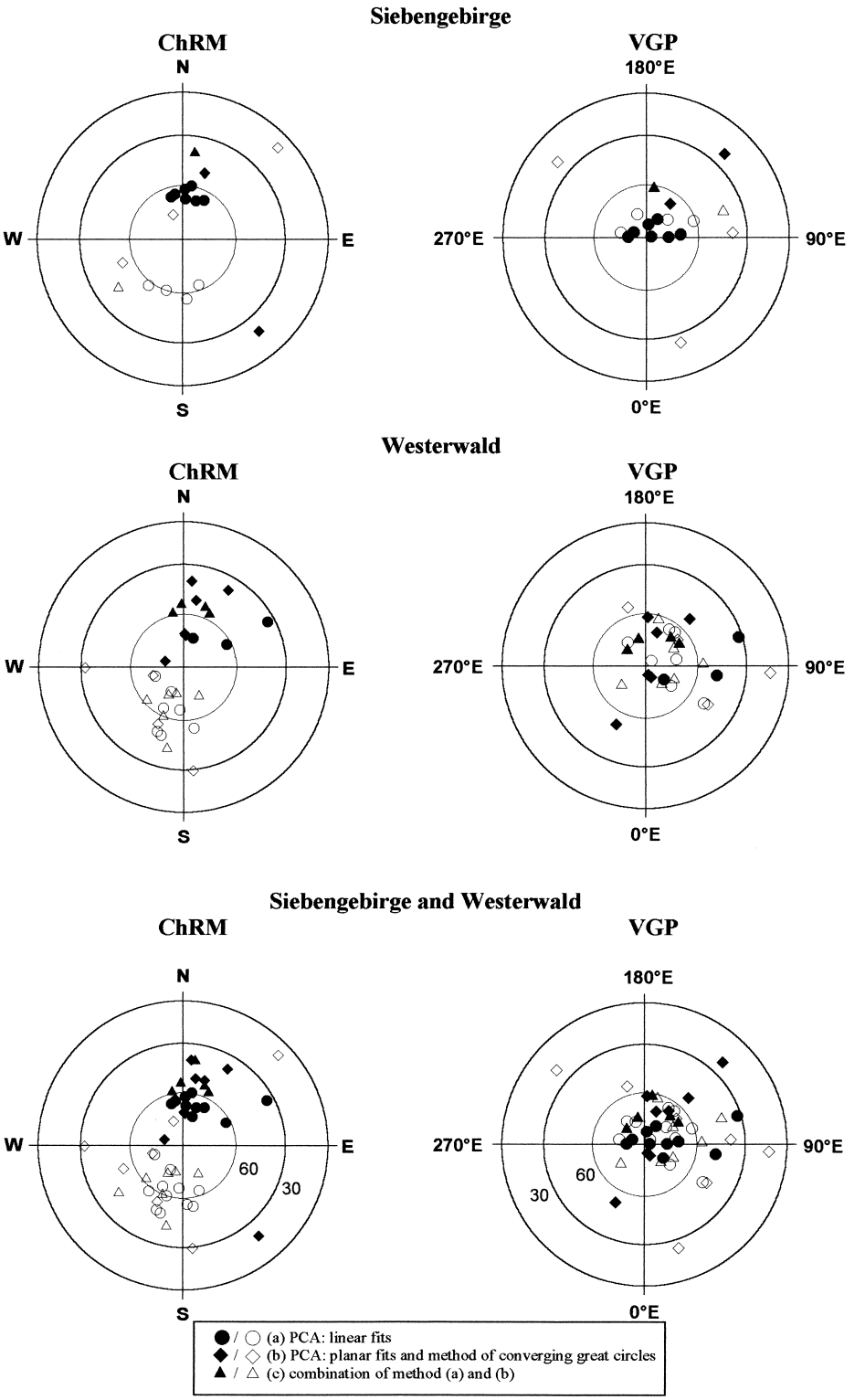
The Westerwald volcanic field in the northeastern Rhenish Massif is characterized by geochemical zonation in the SW–NE direction. Trachytes and phonolites occur solely in the southwest, together with contemporaneous erupted basalts and basanites. The highly evolved volcanics of the southwest are distributed over an elliptical area of more than 200 km^2. The centre of that area contains the most highly differentiated rocks, e.g. phonolites and alkali-rich trachytes. To the northeast, there is an area of more than 500 km^2 containing only basalts and basanites (Fig. 11). Field investigations indicate that the basaltic volcanics of the northeastern area are younger than the basalts and basanites of the southwestern area (Schreiber, 1996). Between the southwestern and northeastern areas almost all intermediate volcanics, e.g. trachybasalts and trachyandesites, occur. The area with intermediate volcanics overlaps the trachytic area in the southwest and the basaltic in the northeast. The distance between the centres of the trachytic area and the basaltic area is more than 15 km (Fig. 11).

The geochemical zonation of the Westerwald volcanics is in contrast to Tertiary volcanic fields such as the Hocheifel (Huckenholz and Büchel, 1988) and Vogelsberg (Ehrenberg, 1986). In the Vogelsberg and the Hocheifel, the younger basaltic lavas completely overlie the area with the evolved volcanics.

The reason for the shift in volcanic activity from southwest to northeast may be the movement of the crust to the southwest during the magmatic activity of an ascending mantle diapir.

4. Tectonic setting of the Upper Rhine graben, Hessian depression, Neuwied basin, Lower Rhine basin, and the Bavarian Pfahl zone

The results of palaeomagnetic measurements suggest a rotation of parts of the northeastern Rhenish Massif by more than 16º, particularly the Siebengebirge and Westerwald regions. However, a close analysis of structural data around the whole block of the northeastern Rhenish Massif (e.g. changes in the direction of fold axes crossing the Rhine river in the northern Rhenish Massif and the dimensions of rotation related basins) indicates a possible rotation of approx. 10º. To outline a kinematic model for the tectonics of the second half of the Cenozoic in central Europe we use published structural data mostly from graben regions (mainly from Schad, 1962; Doebl, 1967; Meiburg, 1982; Meyer, 1994; data from the Geological Survey Nordrhein-Westfalen, 1988;


Siebengebirge
ChRM
N
W
E
S
VGP
180°E
270°E
90°E
0°E
Westerwald
ChRM
N
W
E
S
VGP
180°E
270°E
90°E
0°E
Siebengebirge and Westerwald
ChRM
N
W
E
S
60
30
VGP
180°E
270°E
90°E
0°E
60
30
● / ○ (a) PCA: linear fits
◆ / ◇ (b) PCA: planar fits and method of converging great circles
▲ / △ (c) combination of method (a) and (b)

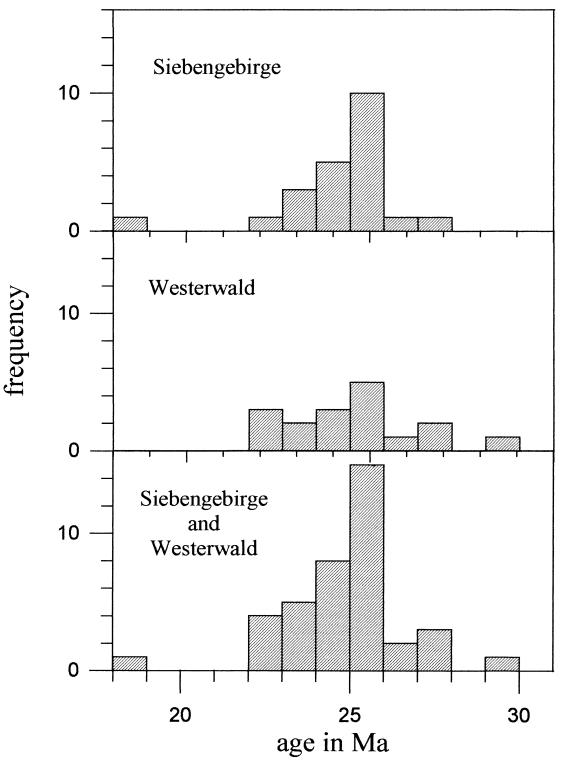


Fig. 8. Age distribution of isotopically dated sites from Siebengebirge and Westerwald after Lippolt and Todt (1978), Todt and Lippolt (1980) and Horn (1993).

data from scale 1 : 25,000 to 1 : 2,000,000 geological maps of the area discussed).

4.1. Upper Rhine graben

The Upper Rhine graben (URG) is about 300 km long and 35–40 km wide and consists of northern and southern segments (Fig. 12). The trend of the graben axis is NNE–SSW, oblique to the NE-trending structures of the Variscan crystalline basement complexes, the Moldanubian in the southern third and the Saxothuringian in the middle and the northern third of the URG. The northern end of the URG is situated to the southeast of the Rhenish Massif on the boundary between the Saxothuringian and Rhenohercynian zones. The basement is uplifted in graben shoulders, the Vosges and Black Forest in the southwest and southeast, respectively, and in the Odenwald in the northeastern shoulder. The boundary between the Moldanubian and Saxothuringian zones is located in the Baden–Baden area in the northern Black Forest and in the Lalaye–Lubine area in the Vosges (Fig. 15). An extrapolation between the Baden–Baden and Lalaye–Lubine areas across the URG shows a sinistral offset of ca. 30 km on a NNE striking fault (Brun et al., 1992). A succession of Lower Permian sediments and volcanics with a thickness of 2–5 km has been deposited on the basement of the northern graben segment. The Permian strata and volcanic complexes in the north and the basement of the remaining area of the Upper Rhine graben are covered by a succession of Mesozoic sediments with thicknesses ranging from 0.5 to more than 1 km. Tertiary sedimentary deposits from Eocene to Recent are between 1800 and more than 3000 m thick (Pflug, 1982). The oldest Tertiary sediments are fine-grained Eocene clastics. They show only little evidence of contemporaneous faulting (Meier and Eisbacher, 1991). However, extension and subsidence within the URG occurred mainly between Oligocene and Miocene time. Graben development itself started by the Early Oligocene in the southern segment documented by synsedimentary faults. Sedimentation shifted to the north over the course of time (Doebl, 1967). Subsidence in the southern segment was interrupted by mantle upwelling induced by the Miocene rift with roughly contemporaneous volcanic activity in the Kaiserstuhl. As a result, the southern graben was uplifted by some 1.5 km and subjected to erosion. In the northern URG near Heidelberg an isolated basin contains more than 1600 m Aquitanian sediments (Schad, 1962). This is interpreted as a pull-apart basin caused by left-hand strike-slip faulting (Schreiber, 1996). Results of the DEKORP-9-N deep reflection seismic investigations of the northern URG suggest that extension below the graben was achieved by slip along a westward-dipping

Fig. 7. ChRM directions and VGP positions obtained from Siebengebirge, Westerwald and Siebengebirge–Westerwald volcanic occurrences. Normal polarity is represented by filled symbols and reversed polarity by open symbols. In case of reversed pole positions the corresponding points on the northern hemisphere are indicated.

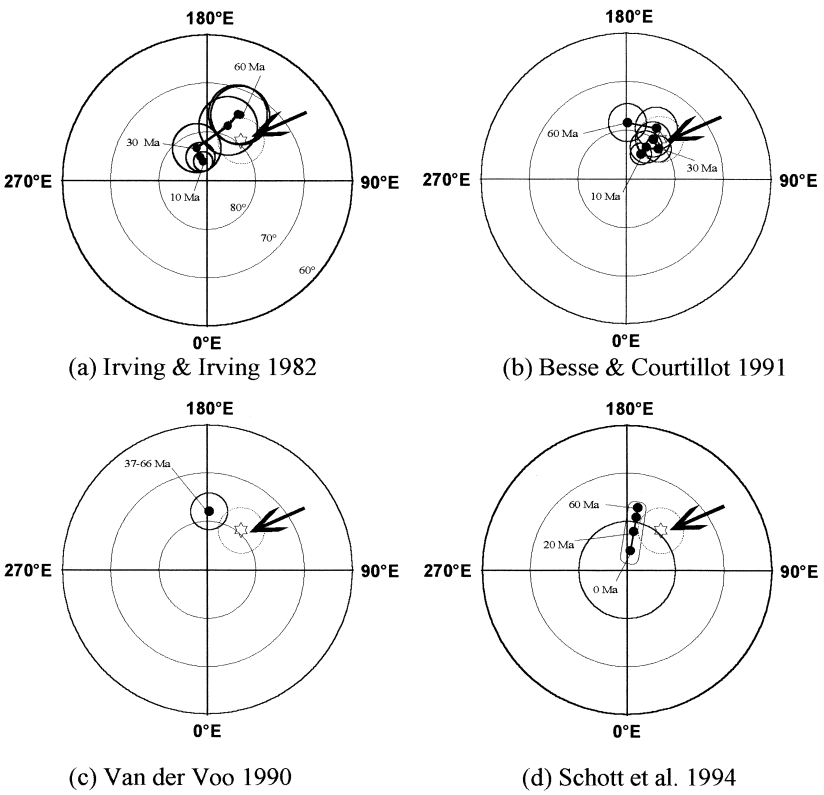


Fig. 9. Comparison of the new Tertiary Siebengebirge–Westerwald palaeopole (open star with dashed α_{95} confidence circle) with different APWP calculations after (a) Irving and Irving (1982), (b) Besse and Courtillot (1991), (c) Van der Voo (1990), and (d) Schott et al. (1994) for Tertiary central Europe. APWP points are represented as dots with solid α_{95} confidence circles. The confidence region for Schott et al. (1994) is schematic, in that work no confidence regions were calculated.

shear zone which continues into the uppermost mantle (Meissner and Bortfeld, 1990). Results of the ECORS/DEKORP deep reflection seismic investigations of the southern URG suggest the opposite: a shear zone dipping gently eastward, which appears to merge at depth with a flat-lying detachment in the ductile lower crust (Brun et al., 1992).

The triangular Mainz basin is situated at the northern end of the URG. The boundary between the basin and the Rhenish Massif to the northwest is defined by the presumably Variscan southern Hunsrück–Taunus fault. To the west, there is a thick succession of Late Carboniferous–Early Permian clastics and volcanics of the Permian Nahe basin. The Mainz basin is a flat sub-basin with a reduced Tertiary sedimentary infill in comparison with the neighbouring northern URG. A first sedimentation of fine-grained clastics took place in the Eocene, as in the whole URG (Doebl, 1967) and other parts of central Europe. Subsidence of the Mainz basin in post-Eocene time was not continuous. Subsidence occurred for a short time in the middle Lattorfian, and after a pause, the main subsidence took place from Rupelian to upper Aquitanian time. In post-Aquitanian time the basin was slightly uplifted together with the Rhenish Massif.

4.2. Hessian depression

One of the main unresolved problems in the Tertiary tectonics of central Europe is the continuation of the URG to the north. At the northern end of the URG the Miocene volcanic complex of Vogelsberg is situated. To the north of the Vogelsberg, a succession of Permian, Mesozoic, and Cenozoic clastics, carbonates, and evaporites some 1000–2500 m thick

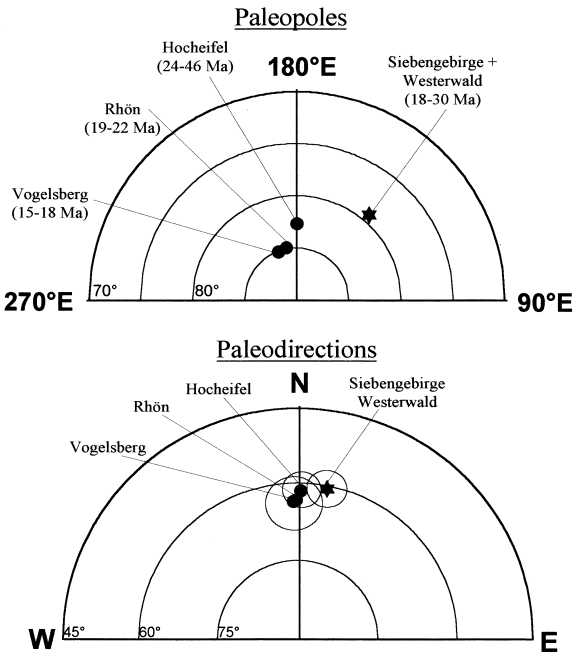


Fig. 10. Comparison of the new palaeomagnetic results from Siebengebirge–Westerwald (asterisk) with other data from the Hocheifel, the Vogelsberg and the Rhön. Palaeodirections were calculated back from palaeopoles for a location with 7º longitude and 50º latitude.

cover the area of the Hessian depression, a broad sag between uplifted parts of the Rhenohercynian, the Harz mountains to the northeast and the Rhenish Massif to the west. In the northern Hessian depression more than 200 m of Upper Permian evaporites have accumulated. Here, salt movement and suberosion processes have sporadically affected tectonic processes. The Hessian depression is framed by extensive graben systems. A graben system striking almost N–S borders the Hessian depression to the west. From north to south, it is subdivided into the Egge, Naumburg–Wolfhagen and Seigertshausen systems. The Altmorschen system is located to the southeast of the Hessian depression and the Leinetal graben to the east (Fig. 17). The eastern and western graben systems converge to the south of the Hessian depression, north of the Vogelsberg volcanic complex. The northwestern border of the Hessian depression is the SW–NE Falkenhagen system and the northeastern border is the NW–SE Elfas–Ahlsburg system. The Hessian depression is divided into individual blocks by some E–W graben systems (Warburg system, Kassel system, Homberg–Lendorfer system). The most conspicuous phenomenon of these E–W systems is the curved course (from northwest to east) of their western part.

Age determination of graben tectonics in the Hessian depression is a major problem. As a result of the uplift of the entire Hessian depression over the last 2 Ma and accompanying deep erosion, Tertiary sediments are scarcely preserved in the graben systems and the border zone of the Hessian depression. Up to now it has generally been assumed that the graben structures are mostly pre-Tertiary. Faulted Tertiary sediments are subordinate (Ritzkowski, 1968). However, almost all indications for the pre-Tertiary

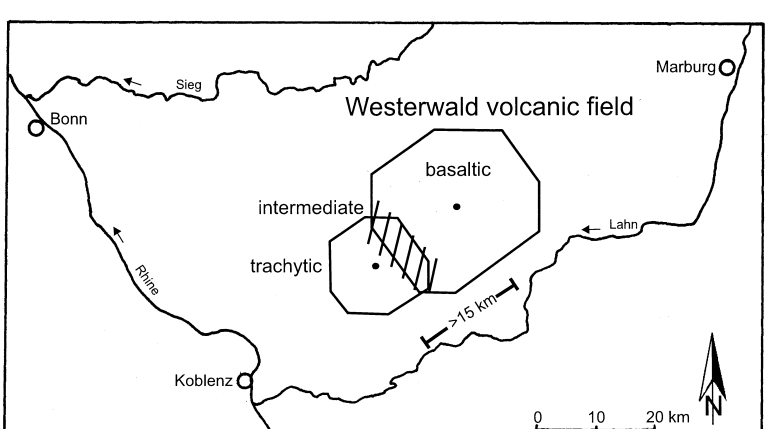
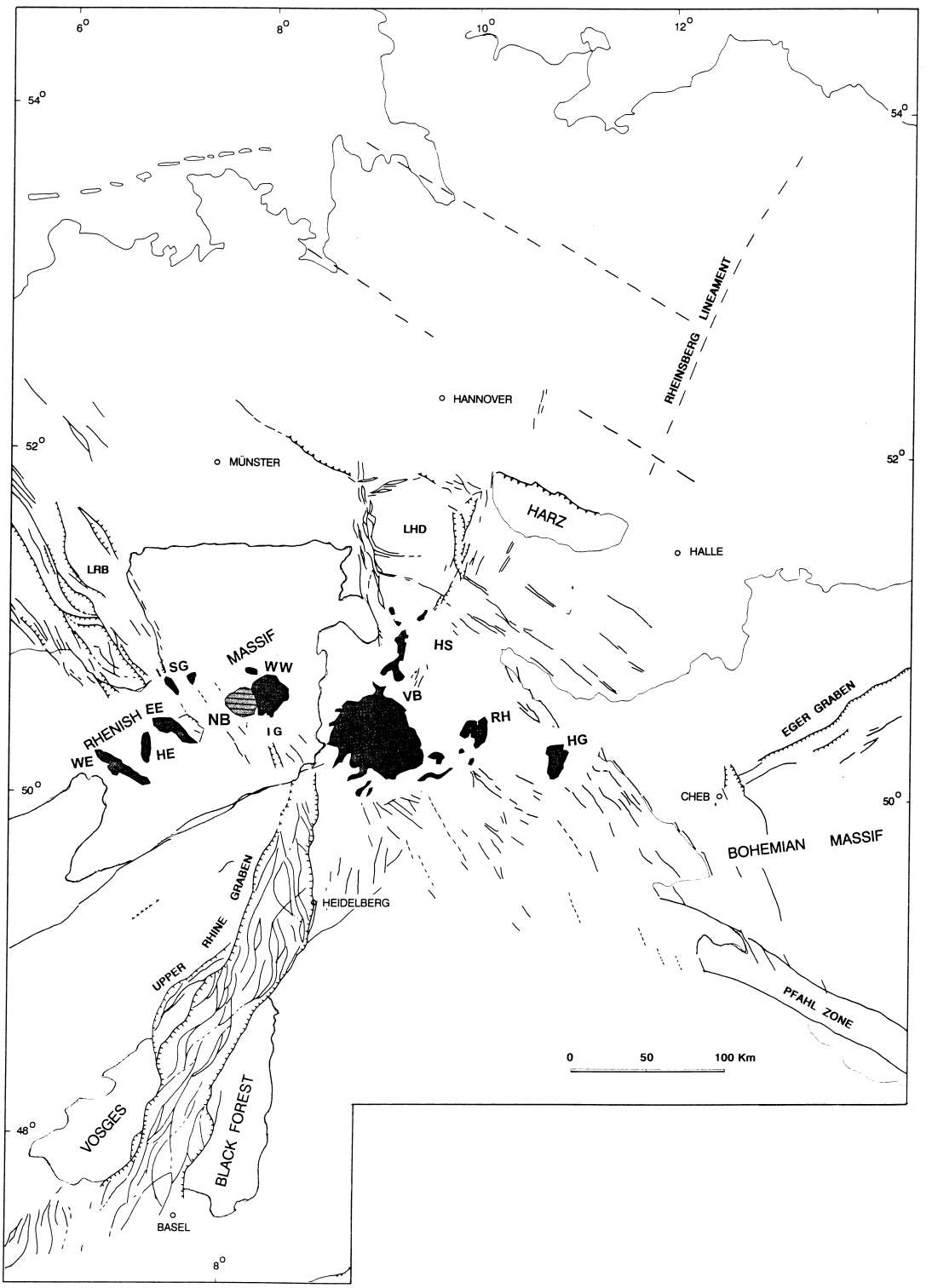


Fig. 11. Geochemical zonation of the Tertiary Westerwald volcanic field. The distance between the trachytic area in the southwest and the basaltic area in the northeast amounts to more than 15 km.


6°
8°
10°
12°
54°
52°
50°
48°
RHEINSBERG LINEAMENT
HANNOVER
MÜNSTER
HARZ
LHD
HALLE
LRB
MASSIF
SG
WW
HS
VB
RHENISH
EE
NB
IG
RH
HG
WE
HE
EGER GRABEN
CHEB
BOHEMIAN MASSIF
UPPER RHINE GRABEN
HEIDELBERG
PFAHL ZONE
0 50 100 Km
VOSGES
BLACK FOREST
BASEL

dating of graben tectonics in the Hessian depression are conclusions drawn by analogy with regions outside the Hessian depression (Stille, 1913; Martini, 1937; Meiburg, 1982). There is no valid indication that the main graben-forming process developed structures before the Tertiary era. In our opinion, the primary tectonic structures of the Hessian depression are Mesozoic fault systems, but the main reactivation and modelling occurred from the Oligocene to the Recent. This is in accordance with investigations made by Rosenfeld (1980) on the Falkenhagen system in the northwestern Hessian depression, by Ritzkowski (1968) on the Seigertshausen system to the southwest, and by Ehrenberg et al. (1992) on the Rhön.

4.3. Lower Rhine basin

From the Netherlands basin to the northwest of the Rhenish Massif, the V-shaped basin of the Lower Rhine basin faces southeast towards the centre of the Rhenish Massif. Sedimentation started in the Early Oligocene in the northwestern part of the basin with a marine succession of sand and clay (Zagwijn, 1989). With the transition from the Early to Late Oligocene, marine transgression spread to the southeast. Here, marine sediments are interspersed with fluvial deltaic deposits, seams of lignite and volcanic tephra from the Siebengebirge volcanic field. The maximum of subsidence and sediment accumulation took place in uppermost Oligocene and lowest Miocene (Aquitanian) time (Hager, 1981), at the same time that more than 1600 m of clastics were deposited in the northern Upper Rhine graben near Heidelberg. In the subsequent Miocene era, subsidence was low and at least 100 m of lignite developed in the central part of the basin without any input of clastics (Hager, 1981). In the Late Miocene, block tilting increased and fluvial deltaic and rare marine sediments were deposited in sub-basins (e.g. Erft block). Fine-grained clastics and lignite are the sediments of the shallow regions. Post-Miocene deposits are dominated by fluvial sand and gravel from the rivers Rhine and Maas (Teichmüller, 1974; Hager and Prüfer, 1988).

The Lower Rhine basin is characterized by a mosaic of northeast tilted blocks which are marked off by NW–SE- and WNW–ESE-trending normal faults, the second with a strike-slip component in some cases. Due to block tilting, several sub-basins and horsts have developed. On the horst blocks, only 200 m of sediments have been deposited whereas the deepest sub-basins contain more than 1200 m of clastics and an intercalated lignite stratum of more than 100 m thickness (Cologne block, Erft block, Rur block). In the northeastern part of the Lower Rhine basin, subsidence ended in the late Tertiary. In the southwestern part of the basin, subsidence of the Cologne block, Erft block, and Rur block has continued up to the present day. Here, more than 100 m of Quaternary fluvial clastics have accumulated (Hager and Prüfer, 1988). The southwestern border zone of the Lower Rhine basin is conspicuously dissected by several NW–SE striking normal faults en echelon. The area of the Lower Rhenish basin has been discussed to be an old Variscan NE–SW shear zone (Loke shear zone, Blundell et al., 1992).

4.4. Neuwied basin

The small isolated basin of Neuwied is located further to the south east of the Lower Rhine basin, at the crossing of the extrapolated flanks of the basin (corresponding to the tip of the V). The oldest sediments of the Neuwied basin are clastics (clay, lignitic sands, and fluvial gravels), presumably from the Upper Eocene (Meyer, 1994) partly comparable with the Eocene sediments of the Upper Rhine graben. In the Oligocene, up to 70 m of predominantly fine-grained clastics were deposited in the Neuwied basin and in the Westerwald area near Montabaur. As a result of an uplift of the whole massif in the Miocene and increasingly in the Quaternary, no large

Fig. 12. Geotectonic map of central Europe with Cenozoic volcanic fields only of the northern Upper Rhine graben area. *WE* = West-Eifel; *HE* = Hocheifel; *EE* = East-Eifel; *SG* = Siebengebirge; *WW* = Westerwald; *VB* = Vogelsberg; *HS* = basalts of the LHD; *RH* = Rhön; *HG* = Heldburg dike swarm; *LHD* = Lower Hessian depression; *LRB* = Lower Rhine basin; *NB* = Neuwied basin; *IG* = Idstein graben. Sources: Geol. Map Hessen, 1 : 300,000; Geol. Map Bundesrepublik Deutschland 1 : 1,000,000; Geol. Map 1 : 200,000: CC 4718 Kassel, CC 5518 Fulda, CC 6318 Frankfurt a.M. East, CC 5510 Siegen; faults of the Upper Rhine graben after Illies (1978).

masses of younger sediments were deposited. On the contrary, it is possible that erosion of Miocene clastics took place. Normal faults of the approximately triangular basin run from NE–SW to NW–SE. A NW–SE right-hand strike-slip fault along the Rhine river near Andernach displaces the NE–SW normal fault. Up to the present day, the deepest part of the basin, at its northern corner, has subsided some 350 m relative to the uplifted surroundings (Bibus, 1983; Meyer, 1994).

4.5. Bavarian Pfahl zone

The Bavarian Pfahl zone is a late Variscan fault system (Bültemann and Hofmann, 1986) on the southwestern border of the Bohemian Massif (Fig. 12). The NW–SE-striking fault is presumably a strike-slip fault which was reactivated as a normal fault, thrust fault and strike-slip fault in the Mesozoic and Cenozoic (Schröder, 1976; Bergerat, 1987). The Pfahl zone is characterized by conspicuous quartz mineralization with up to 100 m thickness in some places. A roughly parallel striking fault is the Danube border fault located about 25–30 km to the southwest. Together with the Pfahl zone these two faults can be seen as a wrench fault system. We assume that the fault system developed at the Vogelsberg in the northwest under the thick cover of Mesozoic strata.

5. Development of stress field in Tertiary time

The horizontal component of maximum compressive stress of the present-day stress field in central Europe has a NW–SE direction (Letouzey, 1986; Bergerat, 1987; Ziegler, 1987; Kooi et al., 1989; Rebaï et al., 1992; Hibsch et al., 1995). Results from fault-plane solutions in central Europe give a mean direction of $\sigma 1_{\mathrm{H}} = 142 \pm 20^{\circ}$ (Ahorner, 1975). Seismotectonic block movements are of the strike-slip type along major fracture zones, striking NNE (Upper Rhine graben) or WNW (Belgian zone after Ahorner, 1975). The seismotectonic block movements are controlled by the stress field, which is caused by interaction between ocean-floor spreading in the North Atlantic Ocean and plate collision in the Mediterranean region.

There has been considerable discussion concerning a change in the direction of maximum compressive stress in the Tertiary (e.g. Illies, 1978; Bergerat, 1987; Meier and Eisbacher, 1991). The axis of maximum compressive stress $\sigma 1_{\mathrm{H}}$ may have changed from NNE–SSW to NW–SE in post-Cretaceous time (Ahorner, 1975). As no brittle deformations are found in Cenozoic sediments, the exact time of the change in stress direction, which may have occurred before the beginning of subsidence of parts of the Rhine graben system or during Rhine graben subsidence, is still unclear. Paleocene basins and grabens are scarce in central Europe and there is no evidence of stress directions at this time. In the Eocene, small isolated basins are assumed to have developed in the whole of the area discussed, accumulating clay from the surrounding peneplains. These basins are preserved in more recent graben structures or under a cover of Eocene/post-Eocene volcanics. Due to the uplift of the whole of central Europe except the graben regions, the primary distribution of Eocene basins is unknown and the precise determination of stress directions is doubtful. The most useful indications come from volcanic dikes and elongations of volcanic fields. The Eocene Hocheifel volcanic field (Lippolt, 1983) is elongated in NNE–SSW direction and volcanic dikes of the presumably Eocene Heldburger Gangschar (age determination is still uncertain; Lippolt, 1982) have the same direction, parallel to the presumed $\sigma 1_{\mathrm{H}}$ direction. From NNE-trending basaltic dikes in the URG border zone, dated at between 55 and 38 Ma, it is assumed that the direction of the early stage of graben extension in Eocene time was WNW–ESE. From isopach trends for the Lower and middle Oligocene, which are parallel to the border faults of the URG (Doebl, 1967), the same direction of extension is assumed (Meier and Eisbacher, 1991). As mentioned above, the main development of the URG itself started in the Early Oligocene (about 35 Ma). This is indicated by coarse clastics derived from uplifted shoulders, which are deposited along the main border faults and contain evidence of contemporaneous faulting. For this time no volcanic dikes are available around the URG as stress indications. Isopach trends only give information on a regional stress field orientation. In view of the fact that the subsidence of the Lower Rhine basin is perpendicular to the URG,

and started slightly later (Rupelian) than the URG, the orientation of $\sigma 1 = S_H$ of the general Oligocene stress field in central Europe could not be NNE–SSW. Therefore, we assume that the maximum horizontal stress direction rotated from NNE–SSW to NNW–SSE in Late Eocene/Early Oligocene time and that rotation continues to NW–SE by the Late Oligocene. In the Late Oligocene, the Siebengebirge volcanic field (Late Oligocene/Early Miocene), with NW–SE elongation, developed.

It is generally accepted that for the last 22 Ma (Aquitanian to Recent) the direction $\sigma 1 = S_H$ has been NW–SE (Meier and Eisbacher, 1991). This is in accordance with the current direction of the greatest horizontal stress S_H (NW–SE) obtained from borehole breakouts, analysis of focal plane solutions, and near-surface in-situ stress measurements (Illies and Greiner, 1978; Ahorner et al., 1983; Mueller et al., 1992; Zoback, 1992). The orientation of $\sigma 1 = S_H$ reflects a transtensional setting for the URG which is accepted for the last 22 Ma. As we will show in the next section we assume that the orientation of $\sigma 1 = S_H$ was NNW–SSE in the Early and middle Oligocene and has been NW–SE since the Late Oligocene, remaining roughly constant for the last 35–30 Ma. This reflects a transtensional setting of the URG with a NNW–SSE direction of greatest horizontal stress S_H in the Early and middle Oligocene and a NW–SE direction of S_H since the Late Oligocene.

6. Seismicity of the Rhine graben system

The Rhine graben system shows recent seismicity in the Lower Rhine basin and related areas, the middle Rhine, and the URG (Ahorner, 1975, 1983; Leydecker, 1986; Camelbeeck et al., 1994), while the Hessian depression has been recently inactive. In the Lower Rhine basin, strike-slip shocks and dip-slip shocks occur in the same order of magnitude. The dip-slip mechanisms are of the tensional type with fault planes trending usually NW–SE. Fault planes of the strike-slip mechanism strike WNW (or NNE). They mainly show a minor dip-slip component with tensional character. The focal point of activity is to the southwest of the Lower Rhine basin and further west in the Belgian earthquake zone. In the Belgian earthquake zone, fault-plane solutions show predominantly dextral strike-slip faulting along WNW–ESE-trending shear zones or sinistral strike-slip faulting along NNE–SSW-trending shear zones. The angle between the main faults of the Lower Rhine basin (about N142ºE, Ahorner, 1975, 1983) and the WNW–ESE-trending shear zones amounts to 30–45º.

Another earthquake zone, the Neuwied basin, is located to the south of the Lower Rhine basin, and further southeast in the middle Rhine area. Fault-plane solutions indicate predominantly left-lateral strike-slip faulting striking NW–SE (about N155ºE), in some locations with a compressional dip-slip component. The strike of the horizontal component of maximum compressive stress (Ahorner et al., 1972; Ahorner, 1983) from strike-slip shocks near Boppard (1970) and Godesberg (1971) is approx. N108ºE and N110ºE, respectively (Ahorner, 1975). This direction of S_H is in contradiction to the direction of S_H (N142ºE) in the Lower Rhine basin (Fig. 16). The angle between the main faults of the Lower Rhine basin (N142ºE) and the assumed shear zones of the middle Rhine (N155ºE) amounts to more than 10º.

In the URG region most fault-plane solutions of earthquakes show strike-slip mechanisms. The relative motion direction parallel to the graben axis is left-lateral with respect to the NNE-trending master faults, indicating a NW–SE compression of the crust. Dip-slip mechanisms are rare and these are of the tensional type indicating extension on NW–SE-trending faults in the basement at an oblique angle to the graben axis (Ahorner et al., 1983; Zippelt and Mälzer, 1987).

7. A kinematic model of the rotational history

Results of the palaeomagnetic study of Neogene volcanic rocks situated within the northeastern Rhenish Massif (northeast of the middle Rhine) suggest that parts of the northeastern Rhenish Massif or the whole Rhenish Massif have undergone a clockwise tectonic rotation of 16º at least since the Late Oligocene. Despite the results of mean rotation values of 16º in Westerwald and Siebengebirge, it seems advisable to assume rotation of only 10º on the basis of geological features (e.g. change of Variscan

fold axes across the northern middle Rhine, dimensions of transrotational basins, etc.). The rotation value of 16° may be the result of rotation of the northeastern Rhenish Massif as a whole, combined with synchronous rotation of discrete micro-blocks in Siebengebirge and Westerwald. In this paragraph we give an interpretation of tectonic structures at the eastern, southern, and western border zones of the rotated block with respect to the rotation kinematics. The exact position of the northern border zone of the rotated block is part of recent investigations (we suppose that it is located to the north of the Cretaceous basin of Münster in the Osning zone) and has been disregarded at this stage of the discussion. In order to develop a kinematic model for the rotational history of the eastern Rhenish Massif, we used published maps in various scales to define first-order strike-slip faults. These faults normally develop in the crustal basement and are covered by younger strata. Small-scale movements at strike-slip faults result in en-echelon second-order faults, thrusts, and folds in the superimposed rock cover which are used for defining them.

From the sedimentary infill of the Lower Rhine basin and URG, the climax of graben and basin subsidence can be identified as the Chattian/Aquitanian (Doebl, 1967; Teichmüller, 1974; Hager, 1981). In the same time span, the magmatic centre of the Westerwald shifted from southwest to northeast (Schreiber, 1996). Following a period of reduced activity in the Middle Miocene, a greater tectonic impulse occurred from the Late Miocene to Recent, with increased uplift of the Rhenish Massif over the last 2 Ma (Meyer, 1994).

The following essential points are part of the kinematic model which is presented:

(1) Development of a conjugate shear system since the Early Oligocene with a left-hand strike-slip master fault in NNE–SSW direction in the URG area and to the north in the region of the Hessian depression with continuation into the basin of Lower Saxony. The conjugate right-hand strike-slip system in WNW–ESE uses old (Variscan/Mesozoic) fault systems such as the Bavarian Pfahl zone.

(2) Movement of western Europe at the URG fault at least 30 km to the southwest with transtensional setting for the URG since the Oligocene.

(3) Rotational movement of the northeastern Rhenish Massif of about 10° in combination with strike-slip movements at the URG fault system.

(4) Due to rotation of the northeastern Rhenish Massif restriction of the Hessian depression and initiation of counterclockwise drag rotations of small crustal blocks. (As a result, continuation of the URG to the north was inhibited.)

(5) Development of a V-shaped basin in the Lower Rhine region due to rotation of the northeastern Rhenish Massif.

A possible scenario for the Cenozoic tectonics of central Europe is as follows.

At the end of the Eocene, the direction of maximum horizontal compressive stress changed from NNE–SSW to NW–SE. In the Early and middle Oligocene a conjugate shear system developed with the maximum horizontal compressive stress trending more to N150°E–N160°E (Fig. 13). With these new stress conditions in the Early Oligocene, the western European crust began sinistral movement to the southwest on a fault system which was initiated in the Rhone, Bresse, Upper Rhine, and Hessian depression areas. The main fault of the URG split up into two branches at the boundary between the Saxothuringian and Moldanubian zone in the area of the future Vogelsberg volcanic field. The two branches developed to the north in the eastern and western border zone of the Hessian depression and further north into the Lower Saxony basin. Contemporaneously, a conjugate dextral system in WNW–ESE direction (ca. N120°E) north of the Alps developed with the reactivation of Permian (?) and Mesozoic fault systems. The main faults of the dextral type are the wrench fault system of the Pfahl zone and the parallel fault zone of the northern Harz border. The northern Harz border fault is interpreted as a dextral strike-slip fault which was active mainly in the Late Cretaceous (König and Wrede, 1988). The wrench fault system of the Pfahl zone which developed at the Vogelsberg under the Mesozoic sedimentary rock cover was offset at least 30 km to the south-southwest along the URG strike-slip fault. The wrench fault system continued to the northwest into the Rhenish Massif on the western side of the URG fault. Several small basins and grabens in the Rhenish Massif with Oligocene fine-grained clastics are related to the postulated wrench fault system (e.g. Idstein graben, Limburg basin, basins of the Wester-

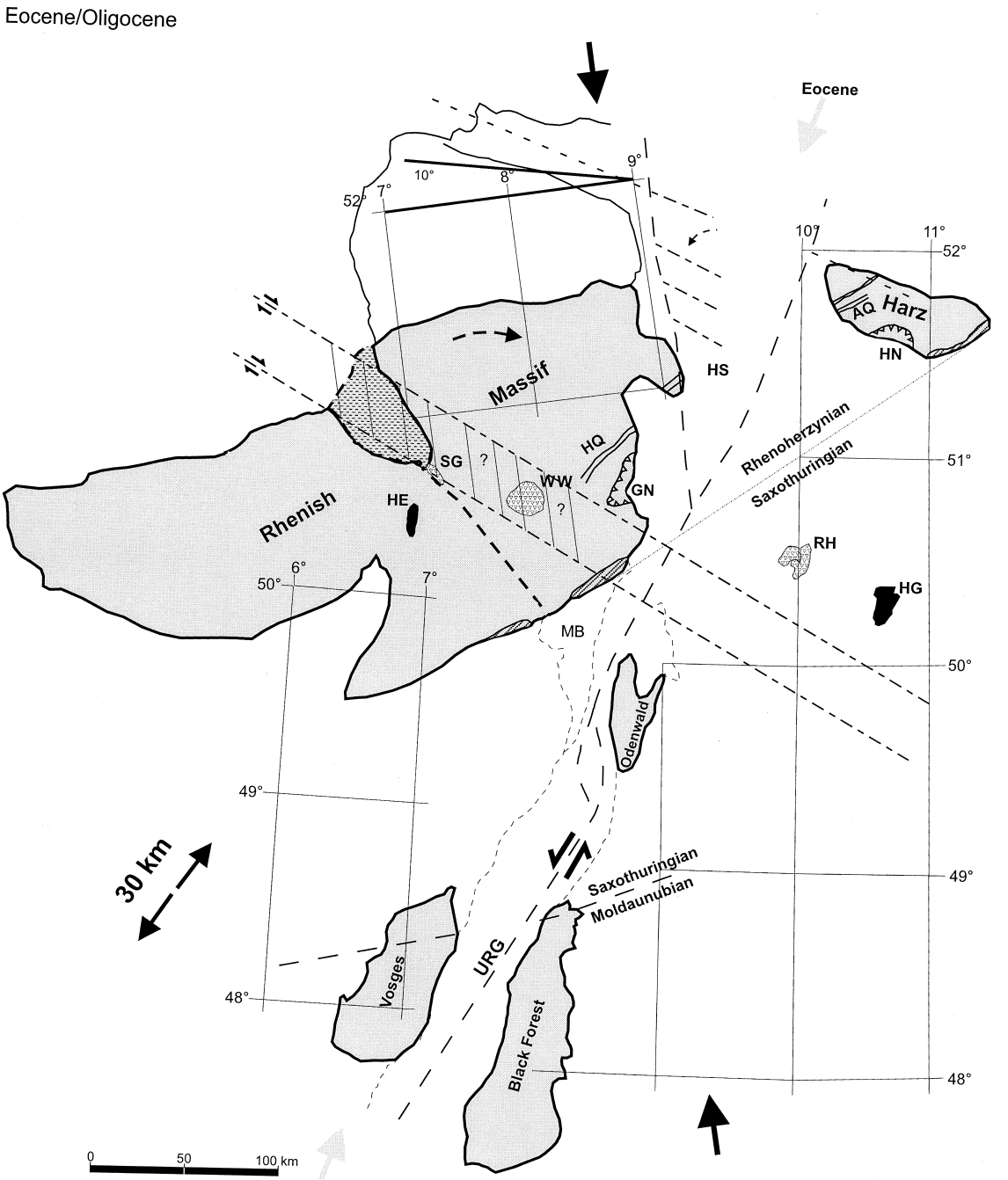


Fig. 13. Palinspastic reconstruction of central Europe for the Eocene/Lower Oligocene and middle Oligocene. The assumed $\sigma 1_H$ direction is NNW–SSE. The northeastern block of the Renish Massif is anticlockwise rotated back by some 10º. The whole of western Europe is removed to the north-northeast by 30 km. Abbreviations: *URG* = Upper Rhine Graben; volcanic fields which existed in Oligocene time: *HG* = Heldburg dike swarm; *HE* = Hocheifel; future volcanic fields: *SG* = Siebengebirge; *WW* = Westerwald; *RH* = Rhön; *HQ* = Hörre quartzite; *AQ* = Acker–Bruchberg quartzite; *GN* = Gießen nap; *HN* = Harz nap; *HS* = Hessian depression; *MB* = Mainz basin.

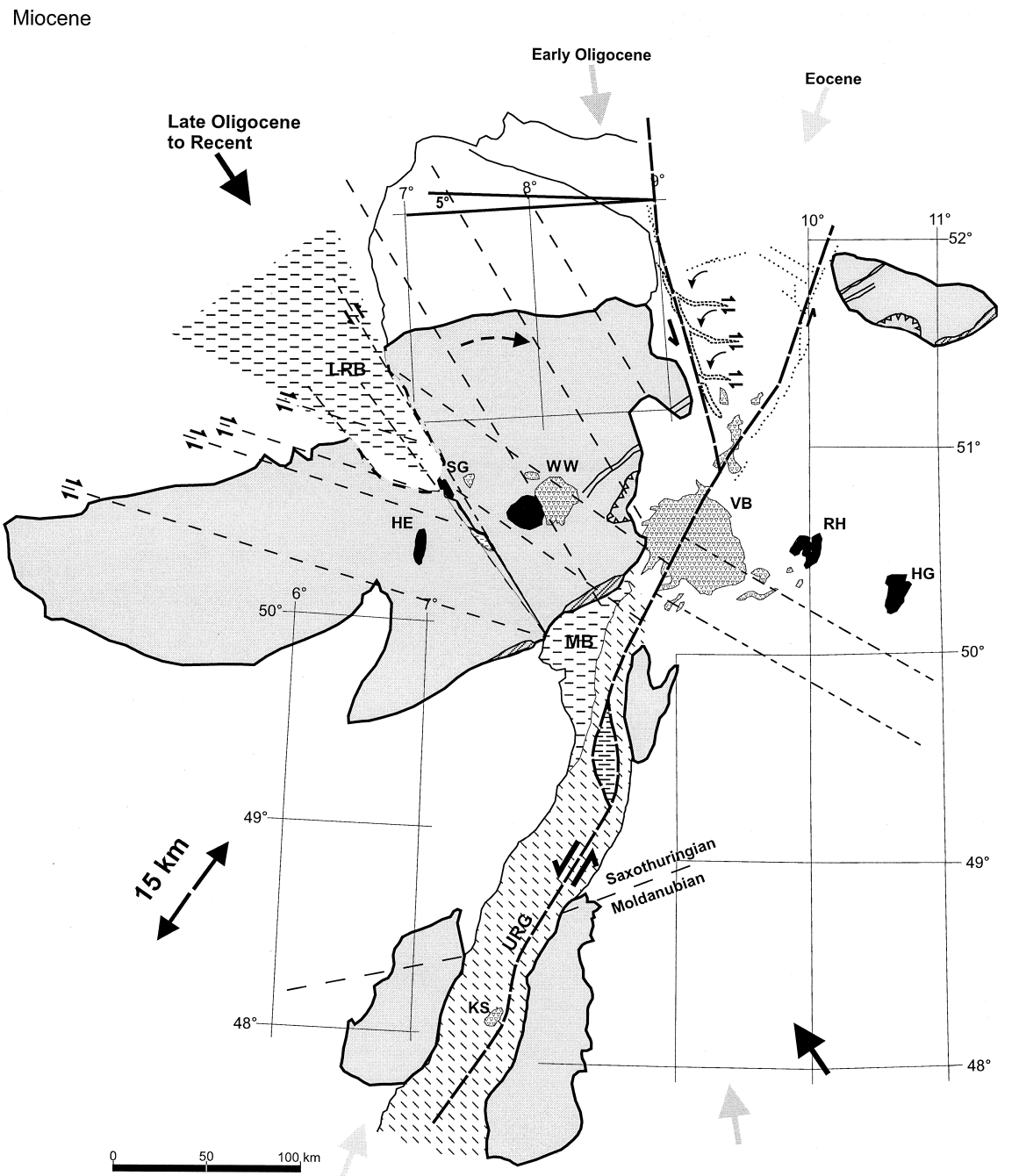


Fig. 14. Palinspastic reconstruction of central Europe for the Miocene. Direction of maximum horizontal stress is NW–SE. Clockwise rotation of the northeastern block of the Rhenish Massif is 5º, sinistral movement of western Europe at the URG fault is 15 km to the southwest. Hatched areas show regions with Tertiary strata. Abbreviations: *VB* = Vogelsberg (Middle Miocene); *KS* = Kaiserstuhl. Other abbreviations as in Figs. 12 and 13.

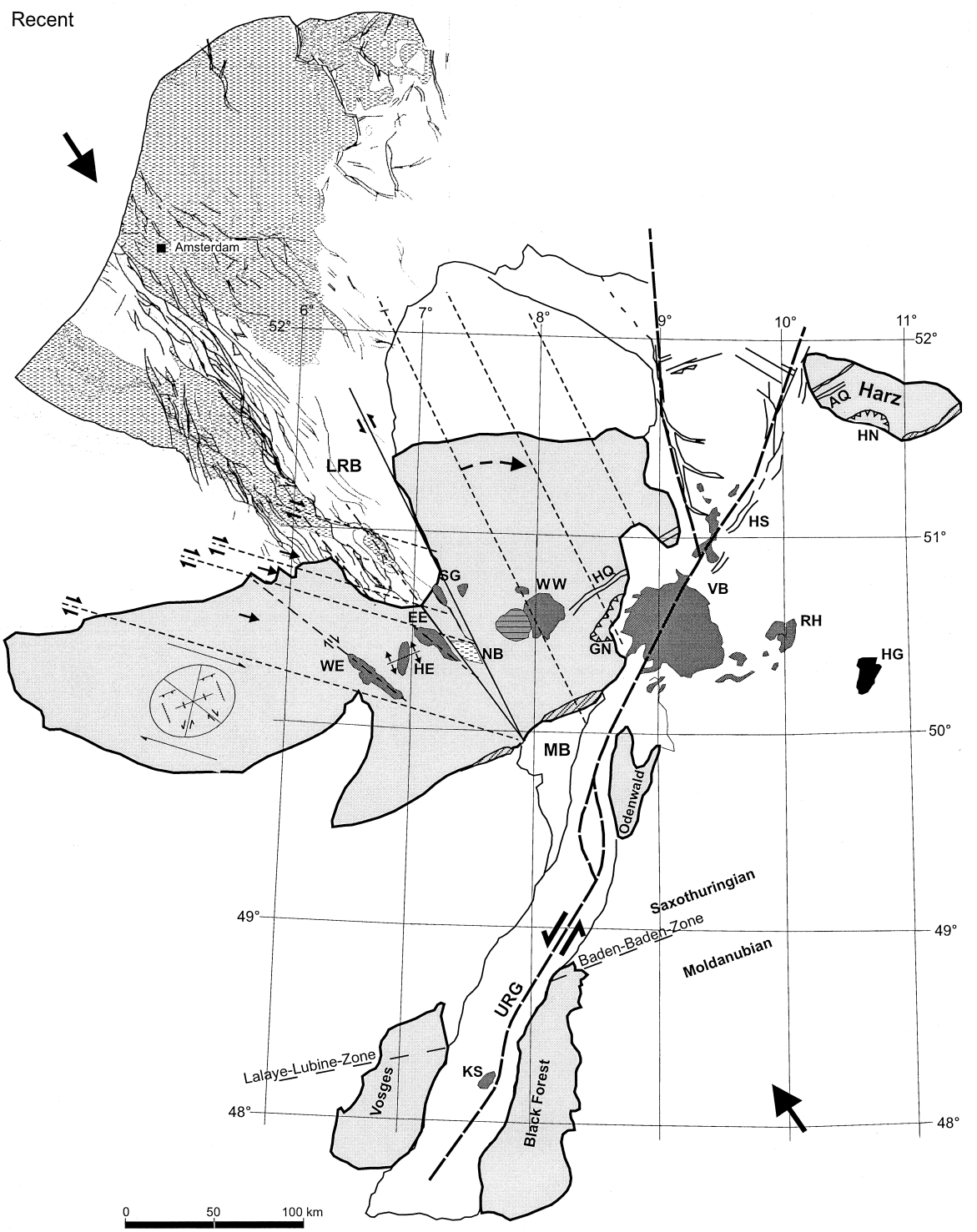


Fig. 15. Tectonic interpretation of the Recent with $\sigma 1_H$ NW–SE direction. For abbreviations see Fig. 13. Hatched areas in the Lower Rhine region show sedimentary deposits with more than 800 m thickness.

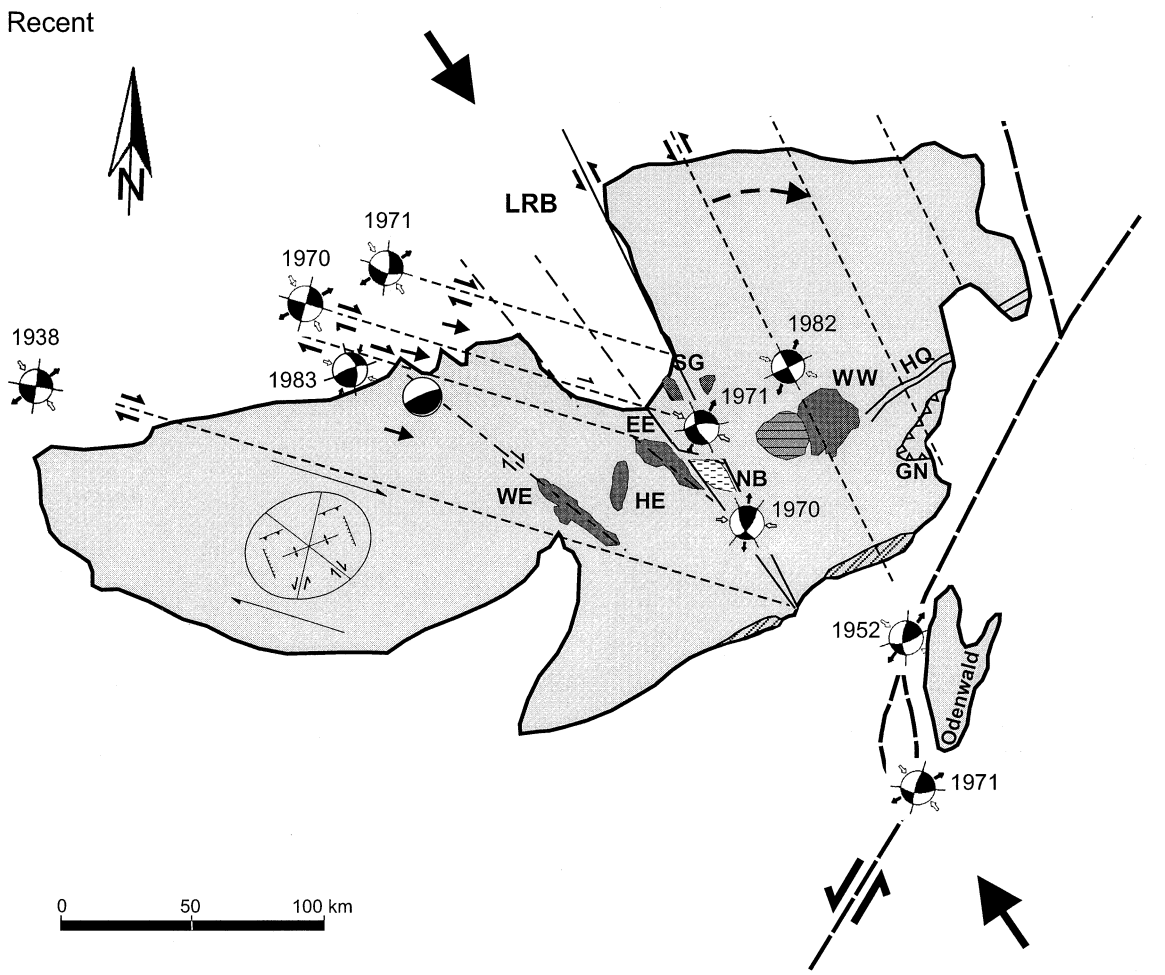
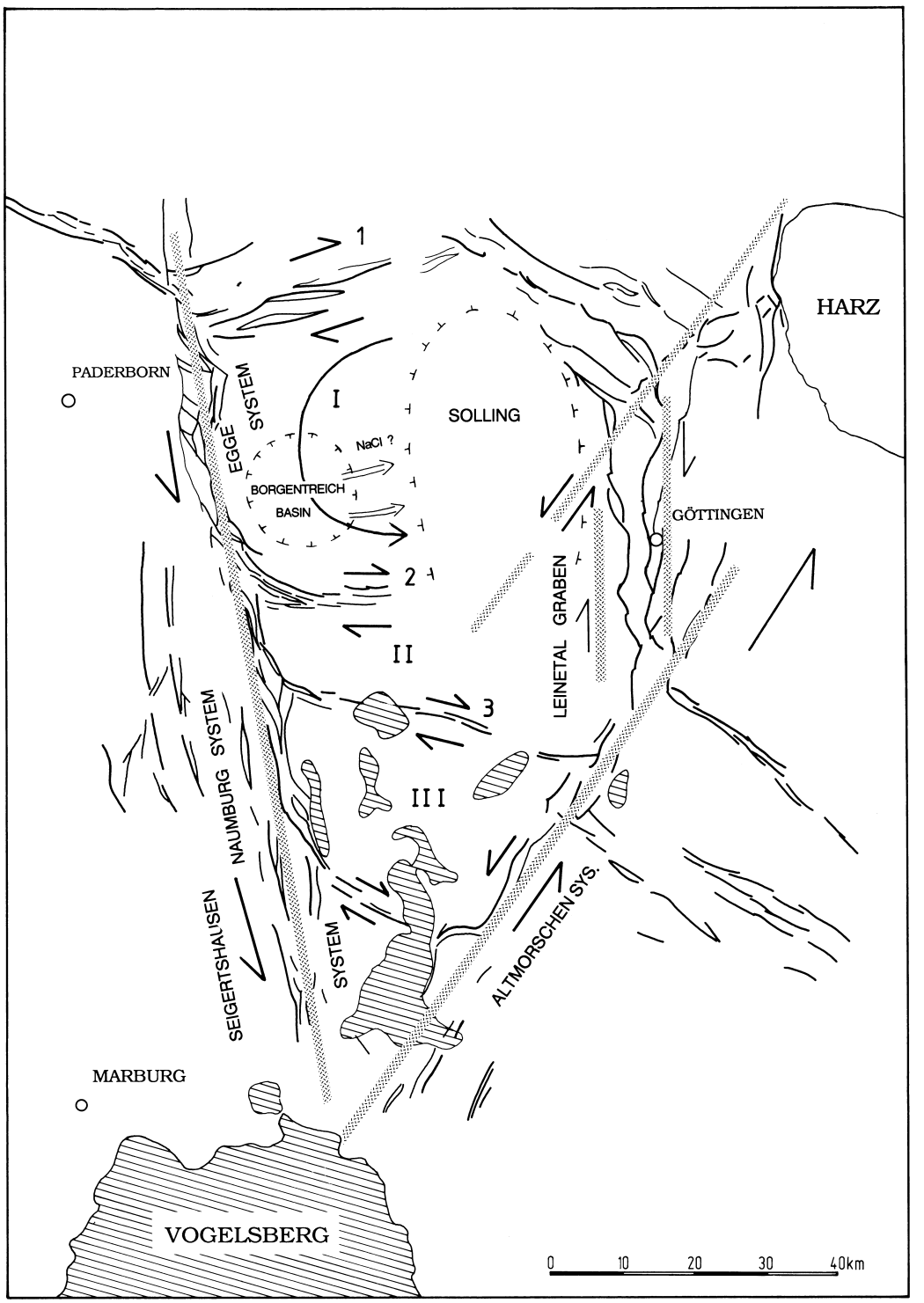


Fig. 16. Focal mechanisms of earthquakes in and around the Rhenish Massif (after Ahorner, 1975, 1983) in combination with the assumed strike-slip faults. Focal mechanisms of earthquakes near Aachen (boarder zone of the western Rhenish Massif to the western Lower Rhine basin) following the 1992 Roermond earthquake show similar fault-plane solutions as the older earthquakes of this region (Prinz et al., 1994). Note the different direction of maximum horizontal stress in the middle Rhine near Godesberg (1971) and Boppard (1970). Fault-plane solution of the Venn earthquakes 1975–1978 (northwest of the West-Eifel, *WE*) show compressional dip-slip dislocation.

wald) and further to NW sub-basins and en-echelon graben structures of the southwestern border of the Lower Rhine basin. The small basins were created by clockwise rotation of NW–SE crustal blocks between the two branches of the wrench fault system (Fig. 13).

In this scenario, the northeastern block of the Rhenish Massif is created in the Late Oligocene by continued sinistral movements at the URG system related to a small change in the direction of maximum horizontal compressive stress to approx. N140°E, as in the Recent. It is assumed that clockwise rotation

Fig. 17. Tectonic interpretation of the Lower Hessian depression (Schreiber, 1994). A split-up of a wrench fault near the Vogelsberg is assumed. *1* = Falkenhagen system; *2* = Warburg system; *3* = Kassel system; *I* = Nethe block; *II* = Zierenberg block; *III* = Istha block; hatched areas: Tertiary volcanic fields of Vogelsberg and the Lower Hessian depression.


HARZ
PADERBORN
EGGE SYSTEM
SOLLING
NaCl ?
BORGENTREICH
BASIN
GÖTTINGEN
LEINETAL GRABEN
NAUMBURG SYSTEM
SEIGERTSHAUSEN
SYSTEM
ALTMORSCHEN SYS.
MARBURG
VOGELSBERG
I
II
III
1
2
3
0 10 20 30 40km

of the block began in the Late Oligocene with the continuing extension of the western European crust to the southwest (Fig. 14). Opening between the rotating and nonrotating crust along the recent Rhine river in the central part of the Rhenish Massif formed both a V-shaped basin in the Lower Rhine basin area and triangular transrotational basins (Ingersoll, 1988; Luyendyk, 1991) in the central part of the Rhenish Massif (e.g. Neuwied basin). The transrotational basins were formed by movements of crustal blocks in the western Rhenish Massif bounded by WNW–ESE dextral faults.

In this model, the tectonics of the Lower Rhine basin were complicated by several superposed effects. Opening between the rotating and nonrotating crust was accompanied by tensional dip-slip faulting with a left-lateral strike-slip component on the northeastern border of the basin. Movement along the WNW–ESE (N120ºE) dextral faults of the wrench fault system related to the older (Early Oligocene) stress direction at the same time as the widening of the Lower Rhine basin led to subsidence of special sub-basins in the southwestern part of the basin and the development of horsts. Second-order oblique dip-slip faults with a dip to the southwest (trending northwest, ca. N140ºE) parallel to the recent axes of maximum horizontal stress were related to the same dextral strike-slip faults of the conjugate shear system. When the maximum horizontal stress reached its present direction in the Late Oligocene, crustal blocks were created in the western Lower Rhenish basin and the western Rhenish Massif bounded by WNW–ESE dextral faults striking N100ºE to N110ºE, resulting in the formation of the Lower Rhenish basin as we know it today (Figs. 14 and 15). In addition, the primary opening fault between the rotating and nonrotating block may have overstepped to parallel faults to the northwest over the course of time. Strike-slip faults with N100ºE to N110ºE directions are found by fault-plane solutions of earthquakes in the Lower Rhine basin (Ahorner, 1975; Prinz et al., 1994; Fig. 16). Strikes of the horizontal component of maximum compressive stress near Boppard and Bad Godesberg are in good accordance with the model of approx. N100ºE fault bounded crustal blocks moving to the southeast of the rotating block. The Quaternary West and East Eifel volcanic fields in the nonrotated western Rhenish Massif are extended in the NW–SE direction (approx. N135ºE). Their position is between the assumed dextral faults striking N100ºE to N110ºE, which is related to synthetic secondary strike-slip faults.

On the northeastern margin of the rotating Rhenish Massif, crustal shortening occurred simultaneously with rotation. In combination with southwest movement of the whole crust at the URG fault system, crustal blocks between the two branches of the URG fault in the area of the Hessian depression rotated anticlockwise. Mesozoic grabens with primary NW–SE directions were rotated almost to an E–W direction (Fig. 17).

At the southern edge of the rotating Rhenish Massif the Mainz basin, which can be seen as a transrotational basin, developed in the Miocene.

Acknowledgements

S. Rotsch is grateful to J. Untiedt (Münster) for helpful discussions. Furthermore he wishes to thank J. Rogenhagen, G. Boyne and H. Stockhausen (all Münster) for help during the field work and N. Petersen and Mrs. D. Körner (München) for support with reflected light microscopy. This work has been supported by the Deutsche Forschungsgemeinschaft (grant: Un 29/39-1). Thanks are extended to W. Meyer (Bonn) for his constructive comments.

References

Ahorner, L., 1975. Present-day stress field and seismotectonic block movements along major fault zones in Central Europe. Tectonophysics 29, 233–249.

Ahorner, L., 1983. Historical seismicity and present-day microearthquake activity of the Rhenish massif, central Europe. In: Fuchs, K., von Gehlen, K., Mälzer, H., Murawski, H., Semmel, A. (Eds.), Plateau Uplift. The Rhenish Shield. A Case History. Springer, Berlin, pp. 198–221.

Ahorner, L., Murawski, H., Schneider, G., 1972. Seismotektonische Traverse von der Nordsee bis zum Apennin. Geol. Rundsch. 61, 915–942.

Ahorner, L., Baier, B., Bonjer, K.P., 1983. General pattern of seismotectonic dislocation and the earthquake-generating stress field in Central Europe between the Alps and the North Sea. In: Fuchs, K., von Gehlen, K., Mälzer, H., Murawski, H., Semmel, A. (Eds.), Plateau Uplift. The Rhenish Shield. A Case History. Springer, Berlin, pp. 187–197.

Bergerat, F., 1987. Stress fields in the European platform at the

time of Africa–Eurasia collision. Tectonics 6 (2), 99–132.
Besse, J., Courtillot, V., 1991. Revised and synthetic apparent polar wander paths of the African, European, North American and Indian plates, and true polar wander since 200 Ma. J. Geophys. Res. 96, 4029–4050.
Bibus, E., 1983. Distribution and dimension of young tectonics in the Neuwied basin and the Lower Middle Rhine. In: Fuchs, K., von Gehlen, K., Mälzer, H., Murawski, H., Semmel, A. (Eds.), Plateau Uplift. The Rhenish Shield. A Case History. Springer, Berlin, pp. 54–61.
Blundell, D., Freeman, R., Mueller, S. (Eds.), 1992. A Continent Revealed — The European Geotraverse. University Press, Cambridge.
Bock, G., Soffel, H., 1971. Paleomagnetic investigations on igneous rocks from the Rhön, Germany. Z. Geophys. 37, 581–587.
Brun, J.P., Gutscher, M.-A., Dekorp-Ecors Teams, 1992. Deep crustal structure of the Rhine Graben from DEKORP-ECORS seismic reflection data: a summary. In: Ziegler, P.A. (Ed.), Geodynamics of Rifting, I. Case History Studies on Rifts: Europe and Asia. Tectonophysics 208, 139–147.
Bültemann, H.W., Hofmann, R., 1986. Die Mineralisation des mittleren Bayerischen Pfahls. Geol. Jahrb. D 83, 3–48.
Camelbeeck, T., van Eck, T., Pelzing, R., Ahorner, L., Loohuis, J., Haak, H.W., Hoang-Trong, P., Hollnack, D., 1994. The 1992 Roermond earthquake, the Netherlands, and its aftershocks. Geol. Mijnbouw 73, 181–197.
Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain size and compositional dependence. Phys. Earth Planet. Inter. 13, 260–267.
Doebl, F., 1967. The Tertiary and Pleistocene sediments of the northern and central part of the Upper Rhine Graben. Abh. Geol. Landesamtes Baden-Württemb. 6, 48–54.
Ehrenberg, K.-H., 1986. Vulkanische Bildungen am Vogelsberg. Fortschr. Mineral. 64 (2), 1–34.
Ehrenberg, K.-H., Hieckethier, H., Rosenberg, F., Strecker, G., Susic, M., Wenzel, G., 1992. Neue Ergebnisse zum tertiären Vulkanismus der Rhön (Wasserkuppenrhön und Kuppenrhön). Beih. Z. Eur. J. Mineral. 4 (2), 47–102.
Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London A 127, 295–305.
Fuchs, K., von Gehlen, K., Mälzer, H., Murawski, H., Semmel, A. (Eds.), 1983. Plateau Uplift. The Rhenish Shield. A Case History. Springer, Berlin.
Geological Survey Nordrhein-Westfalen, 1988. Geologie am Niederrhein. Joh. Van Acken, Krefeld.
Hager, H., 1981. Das Tertiär des Rheinischen Braunkohlenreviers — Ergebnisse und Probleme. Fortschr. Geol. Rheinld. Westf. 29, 529–564.
Hager, H., Prüfer, J., 1988. Geologie am Niederrhein — Tertiär. Geological Survey Nordrhein-Westfalen, Van Acken, Krefeld, pp. 32–40.
Halls, H.C., 1978. The use of converging remagnetization circles in paleomagnetism. Phys. Earth Planet. Inter. 16, 1–11.
Hibsch, C., Jarrige, J.-J., Cushing, E.M., Mercier, J., 1995. Palaeostress analysis, a contribution to the understanding of basin tectonics and geodynamic evolution. Example of the Permian/Cenozoic tectonics of Great Britain and geodynamic implications in western Europe. Tectonophysics 252, 103–136.
Huckenholz, H.G., Büchel, G., 1988. Tertiärer Vulkanismus der Hocheifel. Fortschr. Mineral. 66 (2), 43–82.
Illies, J.H., 1978. Two stages Rhinegraben rifting. In: Ramberg, I.B., Neumann, E.-R. (Eds.), Tectonics and Geophysics of Continental Rifts. NATO Advanced Study Institute Series, Reidel, Dordrecht, pp. 63–71.
Illies, J.H., Greiner, G., 1978. Rhinegraben and the Alpine System. Geol. Soc. Am. Bull. 89, 770–782.
Ingersoll, R.V., 1988. Tectonics of sedimentary basins. Geol. Soc. Am. Bull. 100, 1704–1719.
Irving, E., Irving, G.A., 1982. Apparent polar wander path, Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv. 5, 141–188.
Kirschvink, J.L., 1980. The least-squares line and plane and the analyses of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.
König, S., Wrede, V., 1988. Zur Tektonik der Harzränder. Z. Dtsch. Geol. Ges. 145, 153–171.
Kooi, H., Cloething, S., Remmelts, G., 1989. Intraplate stresses and the stratigraphic evolution of the North Sea Central Graben. Geol. Mijnbouw 68, 49–72.
Letouzey, J., 1986. Cenozoic paleo-stress pattern in the Alpine Foreland and structural interpretation in a platform basin. Tectonophysics 132, 215–231.
Leydecker, G., 1986. Erdbebenkatalog für die Bundesrepublik Deutschland mit Randgebieten für die Jahre 1000–1981. Geol. Jahrb., Reihe E, 36.
Lippolt, H.J., 1982. K/Ar age determinations and the correlation of Tertiary volcanic activity in Central Europe. Geol. Jahrb. D 52, 113–135.
Lippolt, H.J., 1983. Distribution of volcanic activity in space and time. In: Fuchs, K., von Gehlen, K., Mälzer, H., Murawski, H., Semmel, A. (Eds.), Plateau Uplift. The Rhenish Shield. A Case History. Springer, Berlin, pp. 112–120.
Lippolt, H.J., Todt, W., 1978. Isotopische Altersbestimmungen an Vulkaniten des Westerwaldes. Neues Jahrb. Geol. Paläontol. Monatsh. 6, 332–335.
Lowrie, W., Kent, D.V., 1976. Viscous remanent magnetization in basalt samples. Init. Rep. DSDP 34, 479–484.
Lowrie, W., Kent, D.V., 1978. Characteristics of VRM in oceanic basalts. J. Geophys. 44, 297–315.
Luyendyk, B.P., 1991. A model for Neogene crustal rotations, transtension, and transpression in southern California. Geol. Soc. Am. Bull. 103, 1528–1536.
Martini, H.J., 1937. Großschollen und Gräben zwischen Habichtswald und Rheinischem Schiefergebirge. Geotektonische Forsch. 1, 70–123.
McFadden, P.L., Merrill, R.T., McElhinny, M.W., Sunhee, L., 1991. Reversals of the Earth's magnetic field and temporal variations of dynamo families. J. Geophys. Res. 96, 3923–3933.
Meiburg, P., 1982. Saxonische Tektonik und Schollenkinematik am Ostrand des Rheinischen Massivs. Geotektonische Forsch. 62, 1–267.
Meier, L., Eisbacher, G.H., 1991. Crustal kinematics and deep

structure of the northern Rhine Graben, Germany. Tectonics 10 (3), 621–630.
Meissner, R., Bortfeld (Eds.), 1990. DEKORP Atlas. Springer, New York.
Meyer, W., 1994. Geologie der Eifel (3rd ed.). Schweizerbart, Stuttgart.
Mueller, B., Zoback, M.L., Fuchs, F., Mastin, L., Gregersen, S., Pavoni, N., Stephansson, O., Ljunggren, C., 1992. Regional patterns of tectonic stress in Europe. J. Geophys. Res. 97 (B8), 11783–11803.
Nairn, A.E.M., 1962. Paleomagnetic investigations of Tertiary and Quaternary igneous rocks, 1. Preliminary collections in the Eifel, Siebengebirge and Westerwald. Notizbl. Hess. Landesamtes Bodenforsch. Wiesbaden 90, 412–424.
Nairn, A.E.M., Kürsten, M., 1959. Mitteilungen über paläomagnetische Untersuchungen an Basalten des Siebengebirges und seiner Umgebung. Neues Jahrb. Geol. Palaeontol. 8, 348–352.
Pflug, R., 1982. Bau und Entwicklung des Oberrheingrabens. Erträge der Forsch., 184, Wissenschaftliche Buchgesellschaft, Darmstadt.
Prévot, M., 1981. Some aspects of magnetic viscosity in subaerial and submarine volcanic rocks. Geophys. J. R. Astron. Soc. 66, 169–192.
Prinz, D., Hollnack, D., Wohlenberg, J., 1994. The seismic activity near Aachen following the 1992 Roermond earthquake, the Netherlands. Geol. Mijnbouw 73, 235–240.
Rebaï, S., Philip, H., Taboada, A., 1992. Modern tectonic stress field in the Mediterranean region: evidence for variation in stress directions at different scales. Geophys. J. Int. 110, 106–140.
Ritzkowski, S., 1968. Die Tektonik der Braunkohlenlagerstätte von Borken (Tertiär, nördliches Hessen). Notizbl. Hess. Landesamtes Bodenforsch. Wiesbaden 96, 273–280.
Rosenfeld, U., 1980. Der Südwestteil des Lippischen Berglandes. Ein Grenzbereich zwischen Hessischer Senke und Niedersächsischem Tektogen. Z. Dtsch. Geol. Ges. 131, 715–724.
Rotsch, S., 1997. Erdmagnetische Paläopole aus Siebengebirge/Westerwald und die zentraleuropäische Polwanderkurve im mittleren Tertiär. Ph.D. Thesis, Univ. Münster.
Schad, A., 1962. Voraussetzungen für die Bildung von Erdöllagerstätten im Rheingraben. Abh. Geol. Landesamtes Baden-Württemb. 4, 29–40.
Schnepp, E., 1994. Determination of geomagnetic paleointensities from the Quaternary West Eifel volcanic field, Germany. Geophys. J. Int. 116, 688–714.
Schnepp, E., Boyne, G., 1996. Paleointensity measurements of Oligocene volcanic rocks from the Hocheifel (Germany). Ann. Geophys. 13 (Suppl. I), 78.
Schott, J.J., Beck, F., Westphal, M., 1994. Apparent polar wander path smoothing with spherical splines. Phys. Earth Planet. Inter. 85, 67–82.
Schreiber, U., 1994. Die tertiären Vulkanite des Westerwaldes im Rahmen der magmatotektonischen Entwicklung Mitteleuropas. Habilitationsschrift, Univ. Bonn.
Schreiber, U., 1996. Tertiärer Vulkanismus des Westerwaldes. Terra Nostra 96 (7), 187–212.
Schröder, B., 1976. Saxonische Tektonik im Ostteil der Süddeutschen Scholle. Geol. Rundsch. 65, 34–54.
Sherwood, G.J., 1990. A paleomagnetic and rock magnetic study of Tertiary volcanics from the Vogelsberg (Germany). Phys. Earth Planet. Inter. 62, 32–45.
Stille, H., 1913. Die kimmerische (vorcretacische) Phase der Faltung des deutschen Bodens. Geol. Rundsch. 4, 362–383.
Teichmüller, R., 1974. Die tektonische Entwicklung der Niederrheinischen Bucht. In: Illies, J.H., Fuchs, K. (Eds.), Approaches to Taphrogenesis. Schweizerbart, Stuttgart, pp. 269–285.
Todt, W., Lippolt, H.J., 1980. K–Ar age determinations on Tertiary volcanic rocks, V. Siebengebirge, Siebengebirgs-Graben. J. Geophys. 48, 18–27.
Vandamme, D., 1994. A new method to determine paleosecular variation. Phys. Earth Planet. Inter. 85, 131–142.
Van der Voo, R., 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions. Rev. Geophys. 28, 167–206.
Watkins, N.D., 1973. Brunhes epoch geomagnetic secular variation on Réunion Island. J. Geophys. Res. 78, 7763–7768.
Weinreich, N., Bleil, U., 1984. A paleomagnetic study of tertiary volcanics from the Hocheifel (Germany) and the Cenozoic apparent polar wander path of Central Europe. J. Geophys. 54, 163–173.
Wilson, R.L., Dagley, P., McCormack, A.G., 1972. Paleomagnetic evidence about the source of geomagnetic field. Geophys. J. R. Astron. Soc. 28, 213–224.
Zagwijn, W.H., 1989. The Netherlands during the Tertiary and the Quaternary: a case history of coastal lowland evolution. Geol. Mijnbouw 68, 107–120.
Ziegler, P.A., 1987. Late Cretaceous and Cenozoic intra-plate compressional deformations in the Alpine foreland — a geodynamic model. Tectonophysics 137, 389–420.
Ziegler, P.A., 1992. European Cenozoic rift system. Tectonophysics 208, 91–111.
Zippelt, K., Mälzer, H., 1987. Results of new geodetic investigations in SW-Germany. J. Geodyn. 8, 179–191.
Zoback, M.L., 1992. First and second order patterns of stress in the lithosphere: the world stress map project. J. Geophys. Res. 97, 11703–11728.